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Parametric Analysis of a Regenerative H_2 Cycle for Post-Release Buoyancy Stabilization in Heavy-Lift Airships: Actuator Sizing Laws and Scaling Paradox

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Abstract

Heavy-lift airships develop a positive-buoyancy surplus upon payload release that, if unmanaged, drives an uncontrolled ascent. This paper derives actuator sizing laws for a closed regenerative cycle that neutralizes this surplus: H_2 is combusted from the ballonets to reduce net lift; the condensed water is retained as inertial ballast and recovered by on-board solar electrolysis. Three specifications are obtained in closed form and validated across four platforms spanning 5.6–264 t: the minimum combustion rate, which obeys $\dot{m}_{H_2,\mathrm{min}} \propto M_{\mathrm{total}}^{-1/2}$—a *scaling paradox* in which larger craft demand lower specific flow rates; the minimum condenser capacity $Q_{\mathrm{cond,min}} \approx 1{,}512$ kW for a 10-min condensation window; and the minimax-optimal installed rate $\dot{m}^*_{H_2}$, resolved via a depressed cubic equation. Substituting atmospheric air for pure oxygen reduces the descent-phase H_2 budget by $\approx$36% relative to onboard O_2 ($\approx$44.7 vs. $\approx$70 kg/tonne). For the representative platform, anticipatory H_2 extraction of up to $\approx$4 s before each release lowers $\dot{m}^*_{H_2}$ to 5.21 kg/s—an 83% reduction versus a 30 kg/s baseline. Furthermore, a comprehensive mass-budget analysis using 2025 aerospace benchmarks substantiates the hardware feasibility of the proposed cycle: the integrated system requires below 53% of the commercial payload capacity for large-scale platforms, and three of the four evaluated airships achieve energetic self-sufficiency under average irradiance conditions.

Keywords: airship; buoyancy control; regenerative hydrogen; payload release; scaling paradox; anticipatory preconditioning; actuator sizing; solar regeneration

1. Introduction

Airships represent a compelling alternative for long-endurance, low-energy transport in heavy-lift logistics, surveillance, and remote-area supply operations [1,2]. Recent development programs — including the Airlander 10 [3], the Flying Whales LCA60T [4], and the LTA Pathfinder 1 [5] — reflect renewed industrial interest in these platforms for cargo applications at scales unreachable by conventional aircraft. Yet a fundamental operational challenge persists: buoyancy management during payload release. When a load is ejected in flight, the sudden reduction in vehicle mass generates an aerostatic imbalance that can displace the airship by several meters within seconds, with load oscillations during the maneuver posing critical risks to ground personnel and infrastructure [6].

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Traditional responses — venting lifting gas or releasing consumable ballast — resolve the immediate perturbation at the cost of irreversible material loss per delivery cycle, limiting mission autonomy and creating dependency on ground-based resupply. Alternative approaches have been proposed: sliding ballast configurations provide attitude authority but do not address net buoyancy surplus after release [7]; inflatable vacuum chambers offer a closed-cycle alternative but carry prohibitive structural mass penalties that have prevented flight demonstration [8]; and active gas-transfer control has been formulated [2,9], yet existing implementations do not recover the lifting gas consumed during the maneuver. Closed-cycle buoyancy management has been explored through helium compression [10] and regenerative fuel cells [11], but no existing approach integrates dynamic post-release stabilization with on-board H_2 regeneration within a hydrogen combustion–condensation–electrolysis cycle.

This work presents such a cycle. Post-release buoyancy surplus is neutralized by combusting H_2 from the ballonets: the lifting gas is converted into liquid water, which simultaneously acts as inertial ballast and reduces the displaced volume. The water is subsequently recovered as H_2 by on-board solar electrolysis, regenerating the consumed lifting gas for subsequent missions. Substituting atmospheric air for pure oxygen as the oxidizer reduces the descent-phase H_2 demand by $\approx$36% per delivery cycle ($\approx$44.7 kg vs. $\approx$70 kg per tonne with onboard O_2), providing an additional degree of freedom for mission profile optimization.

A key enabler of the cycle's actuator efficiency is an anticipatory preconditioning strategy. Feedforward pre-compensation has been established for suspended-load systems subject to predictable mass perturbations [12], where anticipatory action reduces peak actuator demand and suppresses oscillation buildup. Applied here to the buoyancy domain, a controlled H_2 extraction pulse issued before the scheduled release induces a proactive lift deficit that partially absorbs the perturbation before the reactive control loop engages, shifting the stabilization burden from a purely reactive configuration to a combined feedforward–feedback scheme that substantially reduces the required actuator capacity.

This paper derives the minimum actuator capacity requirements for this cycle across four representative platforms spanning 5.6–264 t. Three specifications are obtained in closed form: the minimum combustion rate $\dot{m}_{H_2,\mathrm{min}}$, which scales as $M_{\mathrm{total}}^{-1/2}$ — a counterintuitive result termed the *scaling paradox*; the minimum condenser thermal capacity $Q_{\mathrm{cond,min}}$; and the minimax-optimal installed flow rate $\dot{m}^*_{H_2}$, resolved via Cardano's method or Vieta's trigonometric substitution depending on the inertia regime (Appendix A). The analysis shows that larger airships are intrinsically easier to stabilize, and that the resulting flow rates constitute directly actionable combustion subsystem specifications. Crucially, a subsystem-by-subsystem mass budget supports the practical engineering feasibility of the system: the hardware envelope operates well within the commercial payload limits of large-scale platforms, and three of the four evaluated airships achieve energetic self-sufficiency under Lima irradiance conditions.

2. Materials and Methods

2.1. System Description and Regenerative Cycle

The regenerative cycle comprises four operationally independent phases, coordinated according to dynamic flight demands. Each phase may be executed asynchronously, without imposing a rigid sequential constraint on system operation (Figure 1).

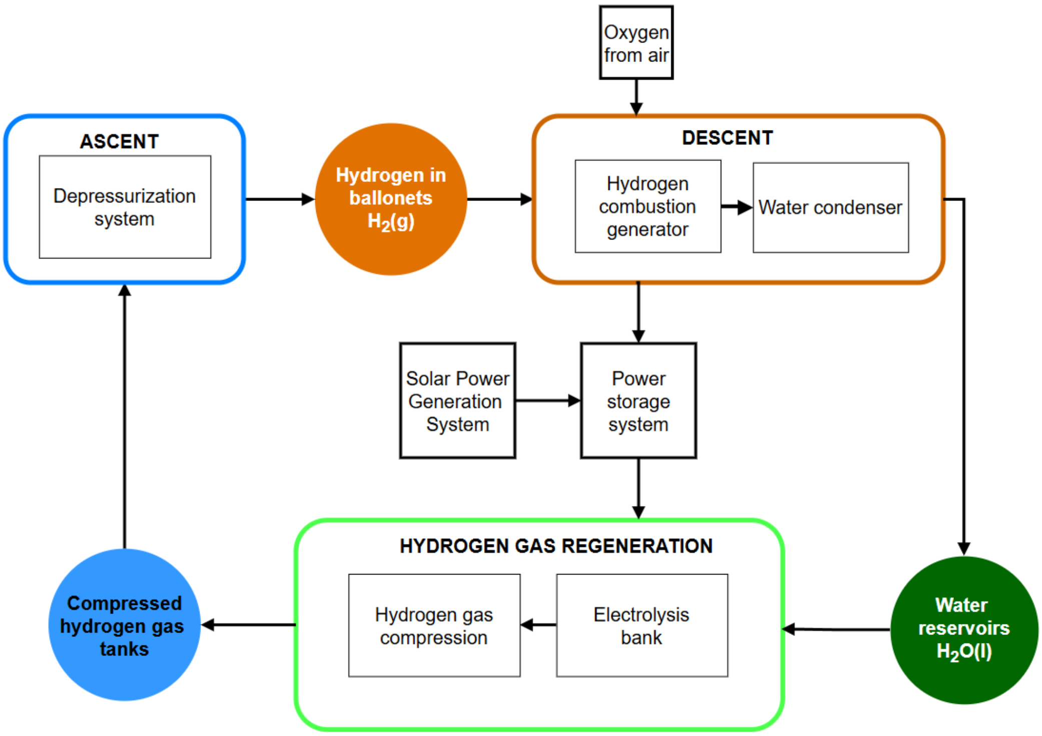

Figure 1. Regenerative H_2 cycle for stabilization after payload release. The blue and orange blocks represent the ascent and descent actuators respectively, both acting on the central state of the system: the H_2 in the ballonets. O_2 is extracted from the surrounding atmosphere, constituting the only external flow in the cycle. Solar regeneration extends operational autonomy but does not participate in the active maneuver.

All fluid-handling and structural subsystems operate in direct or indirect contact with gaseous H_2 and are specified for H_2 service throughout; the envelope and ballonet material architecture is detailed in Appendix C.6.

2.1.1. Ascent Phase

H_2 stored at high pressure (350 bar) is decompressed into the ballonets through the distribution manifold. This inflow increases the displaced volume and, consequently, the net aerostatic thrust governed by Archimedes' Principle.

2.1.2. Descent Phase

To induce altitude loss, a minimum volumetric extraction (on the order of 1 m^3) of H_2 is performed from the ballonets to the generator unit, generating an aerostatic thrust deficit that initiates a controlled descent. At low altitude (10–50 m above the surface), the vehicle enters a regime where aerodynamic drag stabilizes its terminal velocity. Vectorial approach and landing control is handled by the propulsors, operating in a kinematically decoupled manner from the buoyancy system.

2.1.3. Post-Release Stabilization Phase

Post-release stabilization constitutes the central mechanism of the proposed control scheme. Four functionally distinct stages are identified, together implementing the anticipatory transformation of the buoyancy perturbation response:

1. **Proactive preconditioning:** Prior to the maneuver, a fraction of H_2 is preemptively purged from the ballonets, reducing static thrust to mitigate the imminent force imbalance.

2. **Release:** Upon ejection of the payload (Figure 2), the vehicle undergoes a step reduction in its inertial mass, inducing a net upward thrust and a transient vertical acceleration.
3. **Controlled combustion:** The feedback control loop activates the generator unit to oxidize H_2 at a calculated rate that compensates for the aerostatic perturbation. The reference stoichiometric reaction is:

$$H_2(g) + \tfrac{1}{2}O_2(g) \rightarrow H_2O(g), \quad \Delta H = 241\,\text{kJ/mol} \tag{1}$$

generating water vapor with mass ratio $m_{H_2O}/m_{H_2} = 9$. When atmospheric air is used as the oxidizer, N_2 and trace gases act as thermal diluents without participating in the reaction; this effect is captured in the effective efficiency $\eta_{\text{gen}} = 0.30$ adopted in the analysis.
4. **Progressive condensation:** The generated vapor passes through a heat exchanger where it transitions to liquid phase, depositing in the storage tanks. The combustion-condensation cycle achieves an effective volumetric reduction of approximately 1:1300 (calculated stoichiometrically: $\rho_{H_2} \approx 0.083$ kg/m^3 at ambient conditions yields ≈0.75 L of liquid water per m^3 of gas) — each m^3 of gaseous H_2 extracted from the ballonets is consolidated into less than 1 liter of liquid water in the tanks. Since aerostatic thrust depends on the volume of displaced gas rather than the total mass of the vehicle, this conversion effectively reduces net thrust: the water occupies negligible volume compared to the gas it replaces, simultaneously acting as additional ballast.

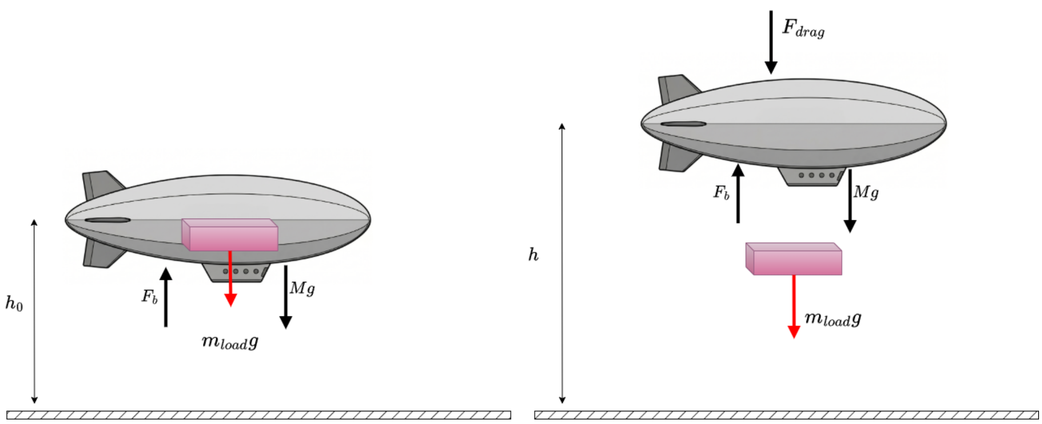

Figure 2. Payload release maneuver of m_{load} from an airship in equilibrium. The excess buoyancy activates the H_2 flow control. F_b denotes aerostatic thrust (Archimedes), Mg the total weight (structure, payload, and gas), and F_{drag} the aerodynamic drag.

2.1.4. Hydrogen Regeneration Phase

The condensed water is subjected to electrolysis to dissociate and recover the consumed H_2(g), which is subsequently recompressed into the main storage tanks. The energy demand of this process is supplied by two redundant sources: the photovoltaic array installed on the outer surface of the airship and the surplus electrical energy generated by the generator unit during the stabilization phase. This latter source recovers combustion energy directly as electrolysis input ($E_{\text{gen}} \approx 447$ kWh per delivery; Section 3.2), covering approximately 22% of the per-delivery electrolysis demand (≈2,012 kWh for 44.7 kg H_2 at 45 kWh/kg); the photovoltaic array supplies the remaining 78%, with self-sufficiency detailed in Section 3.2.

2.2. System Parameters and Justification

Table 1. Model parameters: adopted values and operational rationale.

Parameter	Symbol	Value (Unit)
Control and Dynamics		
Damping coefficient	ζ	1.35
Proactive flow	$\dot{m}_{\text{prec}}$	-10 kg/s
Anticipation time	t_{anticip}	2.0 s
Proportional gain	K_p	15 m^3/m
Derivative gain (nominal)[a]	K_d	2400 $\text{m}^3{\cdot}\text{s}/\text{m}$
Mass control	K_m	60 $\text{kg}/\text{s}/\text{m}^3$
Operation and Flight		
Safety threshold	$\lvert\Delta h\rvert_{\text{max}}$	5.0 m
Steady-state precision	$\lvert\Delta h\rvert_{\text{opt}}$	1.0 m
Initial altitude	h_0	50 m
Max. preconditioning depth	D_{lim}	0.10 m
Thermodynamics and Energy		
Isothermal gas temperature	T_1	288 K
Generator efficiency	η_{gen}	0.30
Electrolysis efficiency	η_{el}	0.65
Electrolysis energy demand	$E_{H_2,\text{el}}$	45.0 kWh/kg
Solar panel fraction	f_{pan}	0.45
Solar irradiance	r_{ef}	2.0 $\text{kWh}/\text{m}^2/\text{day}$

[a] Nominal initial value; recalculated adaptively at each drop via Eq. (A40) (Appendix B).

2.2.1. Justification of Operational and Control Parameters

To ensure the physical viability and industrial applicability of the model, the system parameters were defined based on aerospace regulations and current commercial technology limits (summarized in Table 1).

Flight Dynamics and Control: The altitude controller was tuned by imposing a strict damping coefficient of $\zeta = 1.35$. An overdamped regime is mandatory in external-load operations (such as long slings or aerial cranes) to prevent overshoots and vertical oscillations that would induce fatigue in cargo cables or the dangerous "pendulum effect" described in VERTREP (*Vertical Replenishment*) manuals [13,14]. The proportional-derivative controller gains ($K_p = 15$ m^3/m and $K_d = 2400$ $\text{m}^3{\cdot}\text{s}/\text{m}$ as nominal initial value) were determined to stabilize the maximum structural inertia without saturating the actuators; K_d is subsequently recalculated at each drop event via Eq. (A40) (Appendix B) to maintain $\zeta =$ 1.35 throughout the unloading sequence. A heuristic proactive flow of $\dot{m}_{\text{prec}} = -10$ kg/s was also incorporated, applied 2 seconds before the release. This preconditioning extracts $\approx$20 kg of H_2 (as demonstrated numerically in Section 3), inducing an anticipated lift deficit that absorbs $\approx$44.8% of the initial upward impulse (Eq. (8); buoyancy reduction $\approx$28.8% plus O_2 ballast $\approx$16%) without compromising the airship's altitude. The overdamped tuning criterion is consistent with stationkeeping control strategies reported for large high-altitude airships [15]. Finally, a drag coefficient of $C_D = 0.5$ was adopted, representative of the vertical displacement of large-volume ellipsoidal bodies in turbulent regimes [2,16]. A proportional-derivative structure is selected over integral or state-feedback alternatives for two reasons: (i) the aerostatic system has no persistent steady-state disturbance — the H_2 mass in the ballonets acts as a natural integrator, eliminating altitude error without an explicit I term; and (ii) adaptive gain scheduling handles the system nonlinearity without requiring the full state observer demanded by LQR formulations.

Safety and Mission Thresholds: Simulations begin at an altitude of $h_0 = 50$ m, the typical OGE hovering altitude for aerial crane operations [13]. Two stabilization thresholds were established: a strict safety limit of $|\Delta h| \leq 5$ m, based on the maximum elongation permitted in operational slings to avoid involuntary lifting of ground equipment; and a precision threshold of $|\Delta h| \leq 1$ m, required for structural coupling maneuvers. The vehicle is considered to have reached stabilization (t_{stab}) when, in addition to meeting the spatial threshold, its residual vertical velocity is below 0.05 m/s (0.10 m/s for lighter airships), a safe kinematic margin for interaction with ground personnel [13]. For the multi-release stress test, an interval of $\Delta t = 30$ s was imposed, simulating a high-intensity logistic cadence (colloquially referred to as *carpet-dropping*).

Thermodynamics and Energy Balance: The lifting gas is modeled as isothermal at $T_1 = 288$ K (ISA sea-level standard); temperature deviations during maneuvers of $O(10)$ s are negligible relative to the convective thermal equilibration timescale of large gas envelopes. Hydrogen is supplied from tanks pressurized to 350 bar, the current standard for heavy mobility [17]. The generator efficiency was set to $\eta_{\text{gen}} = 0.30$, a conservative lower bound for hydrogen combustion engines operating on ambient air: comprehensive reviews of H_2 ICEs report peak brake thermal efficiencies of 40–45%, making 30% a robust conservative estimate for continuous off-design operation [18]. For the regeneration cycle, commercial PEM electrolyzers were considered with an energy demand of 45 kWh/kg ($\eta_{\text{el}} \approx 0.65$), consistent with current commercial technology benchmarks [19–21]. Solar recharging assumes a conservative mean irradiance of $r_{\text{ef}} = 2.0\ \text{kWh/m}^2/\text{day}$, characteristic of the central Peruvian coast under stratocumulus cloud cover [22], and a panel coverage fraction of $f_{\text{pan}} = 0.45$, representing the maximum effective area achievable with thin-film solar cells on the upper fuselage curvature [23].

2.3. Model Assumptions and Simplifications

H_2 in the ballonets is modeled as an ideal gas (Eq. (4)), valid for the pressure and temperature ranges considered. Atmospheric O_2 absorption during combustion is explicitly modeled as a dynamic ballast state $m_{\text{water}}(t)$ with stoichiometric rate $\dot{m}_{\text{water}} = -9\min(\dot{m}_{H_2}, 0)$; this state participates in the force balance and the equilibrium volume calculation V_{eq} throughout the simulation. Condensation and electrolysis are treated as decoupled from the flight dynamics, with their timescales evaluated in post-processing (Section 3.2). The control parameters $\dot{m}_{\text{prec}}$, K_p, K_d, and K_m are design values selected through simulation-based tuning to satisfy the overdamped criterion ($\zeta = 1.35$; see Section 2.2). Each model component is grounded in a validated framework: the Newton–Euler dynamics follow Li & Nahon [9], validated against wind-tunnel data for airship platforms; the lifting gas is treated as isothermal at T_1, consistent with the short maneuver timescale ($O(10)$ s) relative to the convective equilibration of large gas envelopes; and the gain-scheduling strategy follows Moutinho et al. [24], applied and validated on real airship platforms. Internal consistency is further confirmed by the agreement between the simulated $\dot{m}_{H_2,\text{min}}$ values and the analytical scaling law (Eq. (11)): three of four platforms agree within 6% of the analytically predicted value (Hindenburg: 2%, Flying Whales: 3%, Pathfinder 3: 6%); the Zeppelin NT achieves a value 25% below the analytical purely-reactive bound, consistent with the greater preconditioning benefit at $\varepsilon = 0.140$.

2.4. Flight Dynamics Modeling

Following the Newton–Euler vertical dynamics framework established for buoyancy-driven airships [9,25], the force balance along the vertical axis is expressed as (Fig. 2):

$$M\,\dot{v} = F_b - M\,g - F_{drag} \tag{2}$$

where $M(t)$ is the total inertial mass (structure, payload, gas in the ballonets, and water ballast accumulated during combustion), F_b the aerostatic thrust (Archimedes' principle, Eq. (3)), and F_{drag} the aerodynamic drag (Eq. (5)). Upon payload release, M undergoes a step reduction while F_b remains instantaneously unchanged, as it depends exclusively on V_{H_2} — a state variable driven solely by $\dot{m}_{H_2}$.

The aerostatic thrust (Archimedes' Principle) depends on air density, which varies exponentially with altitude, and on the volume of H_2:

$$F_b = \rho_{air}(h)\,g\,V_{H_2} \qquad \rho_{air}(h) = \rho_0\,e^{-h/H_s} \tag{3}$$

The gas volume is obtained from the ideal gas law:

$$V_{H_2} = \frac{m_{H_2}\,R_{H_2}\,T_1}{P_{atm}} \tag{4}$$

where $T_1 = 288$ K is the isothermal gas temperature (ISA sea-level standard; temperature deviations during $O(10)$ s maneuvers are negligible; see Section 2.2). The aerodynamic drag force limits the ascent velocity:

$$F_{drag} = \tfrac{1}{2}\,C_D\,A_{front}\,\rho_{air}(h)\,v^2\,\mathrm{sgn}(v) \tag{5}$$

To simulate a multi-delivery mission, the airship mass is modeled as a time-dependent variable. The total inertial mass $M(t)$ decreases in a staircase pattern with each delivery event:

$$M(t) = M_{str} + M_{load}(t) + m_{H_2}(t) + m_{water}(t) \tag{6}$$

where $m_{water}(t)$ is the water ballast accumulated via atmospheric O_2 absorption during combustion ($\dot{m}_{water} = -9\,\min(\dot{m}_{H_2}, 0)$, from the stoichiometric ratio 1:8 H_2:O_2), and the remaining payload $M_{load}(t)$ after $k(t)$ deliveries of mass m_{drop} separated by interval Δt is:

$$M_{load}(t) = M_{load}^{ini} - k(t)\cdot m_{drop}, \quad k(t) = \min(\lfloor t/\Delta t \rfloor, n) \tag{7}$$

where n is the total number of scheduled deliveries. This time-varying nature introduces a direct control challenge: the gains tuned for the initial mass become underdamped at lower masses, generating growing oscillations after consecutive deliveries. The proposed scheme addresses this through two complementary mechanisms: continuous adaptation of the derivative gain and proactive preconditioning prior to each release.

2.4.1. Adaptive PD Controller and Preconditioning

Purely reactive buoyancy control — where the actuator responds only after the perturbation has been initiated — imposes a fundamental constraint on actuator sizing. Upon payload release, the net upward perturbation $\Delta F = m_{load}\,g$ acts on the vehicle immediately; the available response time before the vertical displacement exceeds the safety threshold h_{max} is bounded by $\tau_{ctrl} \approx \sqrt{2\,h_{max}\,M_{total}/(m_{load}\,g)}$ (Eq. (10)). For the smallest evaluated platform (Zeppelin NT, $M_{total} \approx 6.6$ t), this window is approximately 2.6 s. A purely reactive controller must compensate the full perturbation within this interval, implying minimum flow rates of ≈27 kg/s (Eq. (11)) — at the upper boundary of current industrial

H_2 combustion capacity. The hybrid scheme proposed here addresses this constraint by transforming the problem structure *before* the perturbation occurs, following the anticipatory feedforward principle established for suspended-load systems subject to predictable mass events [12].

The stabilization scheme combines a feedback Proportional-Derivative (PD) controller with a proactive preconditioning logic. The PD controller tracks a dynamic equilibrium volume $V_{\text{eq}}(t)$ recalculated at each mass variation. Following the gain-scheduling approach of Moutinho et al. [24], the derivative gain is adaptively tuned to maintain a constant damping coefficient $\zeta = 1.35$ throughout the unloading sequence, guaranteeing a strictly overdamped regime in accordance with VERTREP external-load operational criteria [13,14]. The complete gain expressions and two-phase flow logic are provided in Appendix B.

To mitigate the initial acceleration peak, the system incorporates an anticipatory extraction pulse. Two seconds before the scheduled release instant (t_{drop}), the system temporarily suspends the PD control and injects a baseline constant negative flow of $\dot{m}_{\text{prec}} = -10$ kg/s for the large-scale case studies presented in this work. Extracting this flow for 2 seconds induces a proactive lift deficit sufficient to partially absorb the perturbation before release. With atmospheric O_2 as oxidizer, the absorbed fraction is governed by the effective specific lift factor L_{eff} (Eq. (12)), which combines buoyancy loss and absorbed O_2 mass:

$$\frac{\Delta F}{\Delta F_{\text{pert}}} = L_{\text{eff}} \cdot \frac{|\dot{m}_{\text{prec}}|\, t_{\text{anticip}}}{m_{\text{load}}} \approx 44.8\% \tag{8}$$

for the nominal parameters of Table 1. The buoyancy-only contribution $(\rho_{\text{air}}/\rho_{H_2}) \cdot |\dot{m}_{\text{prec}}|\, t_{\text{anticip}}/m_{\text{load}}$ accounts for $\approx 28.8\%$; the remaining $\approx$16% is provided by the O_2 mass absorbed and retained as water ballast during the 2-second preconditioning window. The anticipation time $t_{\text{anticip}} = 2.0$ s is selected such that the resulting preconditioning sinking depth (Eq. (12)) satisfies $|\Delta h(t_{\text{anticip}})| \approx 0.027$ m for the Pathfinder 3 — negligible relative to the post-drop safety threshold ($|\Delta h|_{\max} = 5$ m) — while remaining well within the operational bound $D_{\text{lim}} = 0.10$ m adopted for the minimax optimization (Table 1).

The hybrid control law with saturation is defined as:

$$\dot{m}_{H_2}(t) = \begin{cases} \dot{m}_{\text{prec}} & \text{if } t \in [t_{\text{drop}} - t_{\text{anticip}},\ t_{\text{drop}}) \\ \text{sat}_{\dot{m}_{H_2,\max}}(\dot{m}_{\text{PD}}(t)) & \text{otherwise} \end{cases} \tag{9}$$

This saturation at $\pm\dot{m}_{H_2,\max}$ is the critical design parameter: if the hybrid scheme demands more flow than the system can provide, the vertical displacement exceeds the operational safety threshold. The proactive pulse qualitatively transforms this requirement: instead of requiring the actuator to compensate for an already-initiated upward impulse, it anticipates the perturbation by reducing net buoyancy *before* the release. The PD controller then acts on a system that is already partially in descent, drastically reducing the reactive flow required.

2.5. Theoretical Analysis of Minimum Requirements

From the linearized dynamics at the equilibrium point (h_0, T_1, P_{atm}) a lower bound is obtained for the minimum actuator capacity: the rate below which no control scheme can guarantee stabilization within $h_{\max}$. Upon payload release, the net perturbation is $\Delta F = m_{\text{load}}\, g$. The thrust sensitivity to mass flow rate, from Eq. (4), is $(\rho_{\text{air}}/\rho_{H_2})\, g$. Without control, the available time before exceeding $h_{\max}$ is:

$$\tau_{\text{ctrl}} \approx \sqrt{\frac{2\,h_{\max}\,M_{\text{total}}}{m_{\text{load}}\,g}} \tag{10}$$

where $M_{\text{total}} - m_{\text{load}} \approx M_{\text{total}}$ has been applied under the assumption $\varepsilon = m_{\text{load}}/M_{\text{total}} \ll 1$. This holds well for three platforms ($\varepsilon \leq 0.010$, Table 2); for the Zeppelin NT ($\varepsilon = 0.140$), the approximation introduces a $\sim$7% underestimate of $\dot{m}_{H_2,\min}$, acceptable for the purposes of this lower-bound derivation.

Imposing that the actuator compensates ΔF within τ_{ctrl}:

$$\dot{m}_{H_2,\min} \geq \frac{\rho_{H_2}}{\rho_{\text{air}}}\sqrt{\frac{m_{\text{load}}^3\,g}{2\,h_{\max}\,M_{\text{total}}}} \tag{11}$$

The result is counterintuitive: $\dot{m}_{H_2,\min}$ **decreases** with M_{total} — larger airships require less actuation capacity, not more. A high inertial mass implies a small perturbation index $\varepsilon = m_{\text{load}}/M_{\text{total}}$, slow post-release acceleration and more available time to respond. The parametric sweep quantifies this trend across four representative cases.

2.6. Analytical Bound Under Hybrid Preconditioning

The reactive lower bound of Section 2.5 assumes the actuator responds exclusively *after* the perturbation has occurred. In the hybrid scheme, the proactive extraction pulse transforms the problem before the release: H_2 exits the ballonets while the vehicle simultaneously absorbs atmospheric O_2 for combustion, yielding a net downward force that grows linearly in time. Applying Newton's second law to this forcing and integrating twice yields a cubic displacement profile (Appendix A):

$$|\Delta h(t)| = \frac{L_{\text{eff}}\,g\,|\dot{m}|}{6\,M_{\text{total}}}\,t^3, \qquad L_{\text{eff}} = \frac{\rho_{\text{air}}}{\rho_{H_2}} + 8 \tag{12}$$

where L_{eff} is the effective specific lift factor that accounts for both the lost buoyancy and the atmospheric O_2 mass absorbed during combustion (8 kg per kg of H_2 consumed, from stoichiometry). The cubic exponent arises because a constant extraction rate produces a linearly growing net force, which integrates twice to a position profile cubic in time.

Since the preconditioning and post-release stabilization phases share the same combustion hardware, the installed actuator capacity $\dot{m}_{H_2,\max}$ must satisfy the peak demand of whichever phase is more intensive. The optimal distribution of the compensation burden between both phases is formulated as a minimax problem (Appendix A):

$$\dot{m}^*_{H_2} = \min_{\alpha\in[0,1]}\Big(\max\big(\dot{m}_{\text{pre}}(\alpha),\ \dot{m}_{\text{post}}(\alpha)\big)\Big) \tag{13}$$

where α is the proactive compensation fraction, $\dot{m}_{\text{pre}}(\alpha) \propto \alpha^{3/2}$ is monotonically increasing, and $\dot{m}_{\text{post}}(\alpha) \propto (1-\alpha)$ is monotonically decreasing. The global minimum of the peak demand occurs at their intersection α^*, which reduces to a depressed cubic polynomial with a unique physical root. This root is resolved in closed form via Cardano's method in the high-inertia regime ($\Delta > 0$) and Vieta's trigonometric substitution in the low-inertia regime ($\Delta \leq 0$) — full derivations in Appendix A. The resulting minimum required flow preserves the $M_{\text{total}}^{-1/2}$ scaling of Eq. (11):

$$\dot{m}^*_{H_2} \propto \frac{m_{\text{load}}^{3/2}}{\sqrt{M_{\text{total}}}} \tag{14}$$

Figure 3 maps $\dot{m}^*_{H_2}$ as a function of the maximum allowed preconditioning sinking depth D_{lim} for the four evaluated platforms. For the minimax-optimal validation of Section 3.1.3, $D_{\text{lim}} = 0.10$ m is adopted as a conservative operational bound: it satisfies

$D_{\text{lim}} \ll |\Delta h|_{\text{max}} = 5$ m (2% of the safety threshold) while providing sufficient preconditioning depth for meaningful compensation. At this operational depth, the analytical minimax bound lies within 15–25% of the numerically simulated values reported in Table 3; the residual gap is attributable to aerodynamic drag and finite-mass effects not captured in the drag-free analytical model.

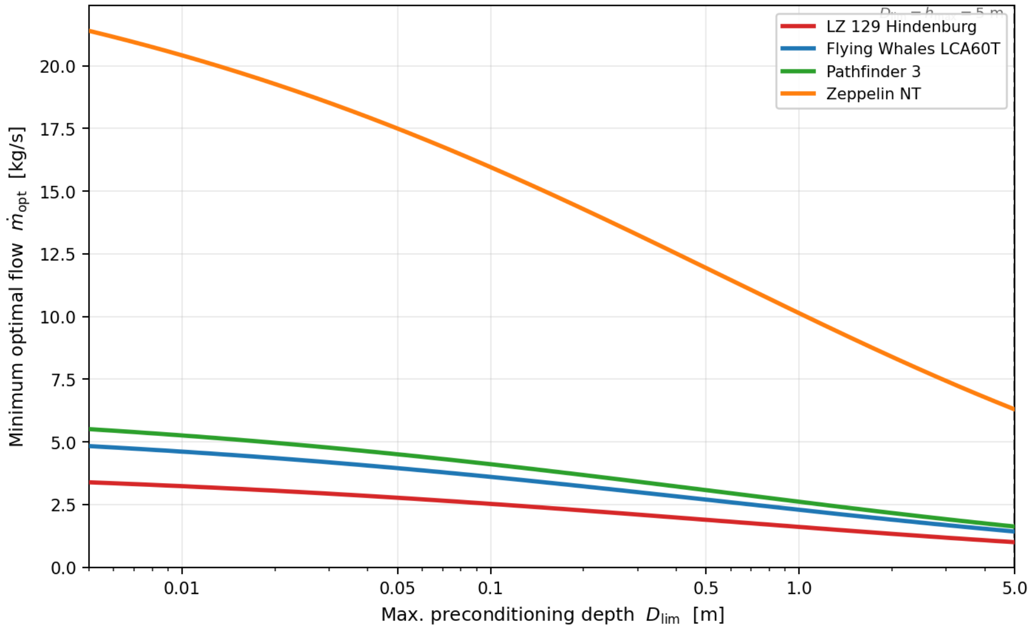

Figure 3. Minimax-optimal H_2 combustion rate $\dot{m}^*_{H_2}$ as a function of the maximum allowed preconditioning depth D_{lim} (Eq. (13); derivations in Appendix A). Each curve corresponds to one evaluated airship. The $M_{\text{total}}^{-1/2}$ scaling is directly visible: larger platforms achieve lower installed capacity requirements at every D_{lim}. The right boundary ($D_{\text{lim}} = 5$ m) coincides with the operational safety threshold h_{max}; the working point of the numerical simulations (Section 3) lies in the low-D_{lim} region of each curve.

3. Results

3.1. Numerical Simulation and Representative Cases

The four platforms were selected to span the full range of structural masses documented for current and projected heavy-lift airships (5.6 t to 264 t), with publicly available technical specifications enabling reproducible parameter assignment. Together they represent the principal scale classes of interest for commercial heavy-lift logistics: ultralight multi-mission (Zeppelin NT), medium heavy-lift (Pathfinder 3), large-capacity (Flying Whales LCA60T), and historical reference of maximum scale (LZ 129 Hindenburg).

Four airships are evaluated with common parameters: $m_{\text{load}} = 1000$ kg, $h_0 = 50$ m (typical OGE hovering altitude for aerial crane operations [13]) and $r_{\text{ef}} = 2.0$ kWh/m²/day (conservative irradiance for the average cloud cover on the Peruvian coast). A_{front} corresponds to the elliptical projection of the airship body on the horizontal plane, the reference area for drag in vertical motion. M_{str} includes structure, systems, and operational fuel load.

Table 2. Characteristics of the evaluated airships. The Payload column lists the nominal commercial cargo capacity reported by each platform's reference source, used as the denominator of the mass-budget envelope of Table 7.

Airship	A_{front} (m²)	M_{str} (kg)	A_{total} (m²)	**Payload** (kg)	ε
LZ 129 Hindenburg [2]	7,950	263,600	24,000	12,500	0.004
Pathfinder 3[a] [5]	4,390	99,000	13,000	10,000	0.009
Flying Whales LCA60T [4]	7,854	129,000	23,000	60,000	0.007
Zeppelin NT [2]	837	5,622	2,800	1,000	0.140

[a] Preliminary specifications from LTA Research; values are estimates.

3.1.1. Parametric Sweep of $\dot{m}_{H_2,\max}$

A sweep of $\dot{m}_{H_2,\max}$ was performed between 0.5 and 1000 kg/s for each airship. For the Zeppelin NT, resolution was increased in the interval $[20, 50]$ kg/s to precisely capture the crossing of the safety thresholds. Throughout the sweep, the preconditioning pulse is bounded by the evaluated actuator capacity — $|\dot{m}_{\text{prec}}| \leq \dot{m}_{H_2,\max}$ — so that at rates below 10 kg/s the preconditioning is proportionally reduced by saturation; the reported $\dot{m}_{H_2,\min}$ values are therefore conservative bounds achievable even under partial preconditioning. The maximum vertical displacement $|\Delta h|$ and stabilization time t_{stab} were evaluated (considered reached when the residual vertical velocity falls below 0.05–0.10 m/s, a safe threshold for ground personnel). The peak acceleration $a_{\max}$ is characterized analytically in Section 4.3.

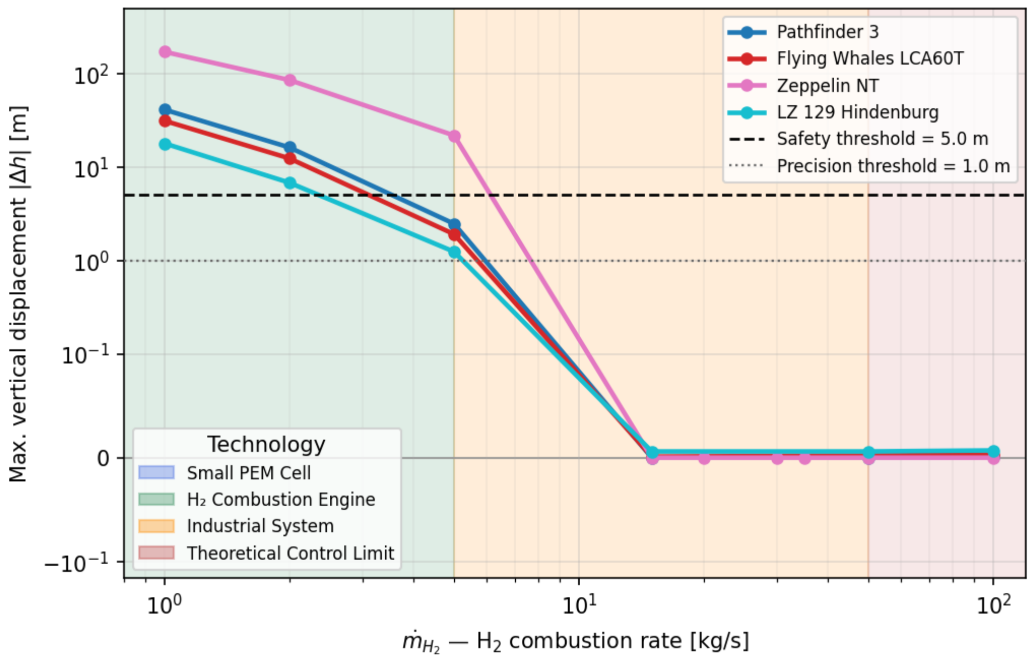

Figure 4. Parametric sweep: maximum vertical displacement $|\Delta h|$ as a function of the H_2 combustion rate $\dot{m}_{H_2}$ for each evaluated airship, accounting for O_2 mass absorption from atmospheric air during combustion. The safety (5 m, dashed) and precision (1 m, dotted) operational thresholds are indicated. Colored background regions classify the combustion rate by applicable technology: H_2 combustion engine ($\lesssim$3 kg/s), industrial system (3–20 kg/s), and theoretical control limit ($\gtrsim$20 kg/s). The crossing of each curve below the 5 m line defines $\dot{m}_{H_2,\min}$ for that platform; the inverse scaling with M_{total} is directly visible (Hindenburg and Flying Whales cross at lower rates than the Zeppelin NT). At high combustion rates, $|\Delta h|$ becomes slightly negative for large-scale platforms: the proactive preconditioning pulse initiates a controlled descent before the release event can induce any net ascent.

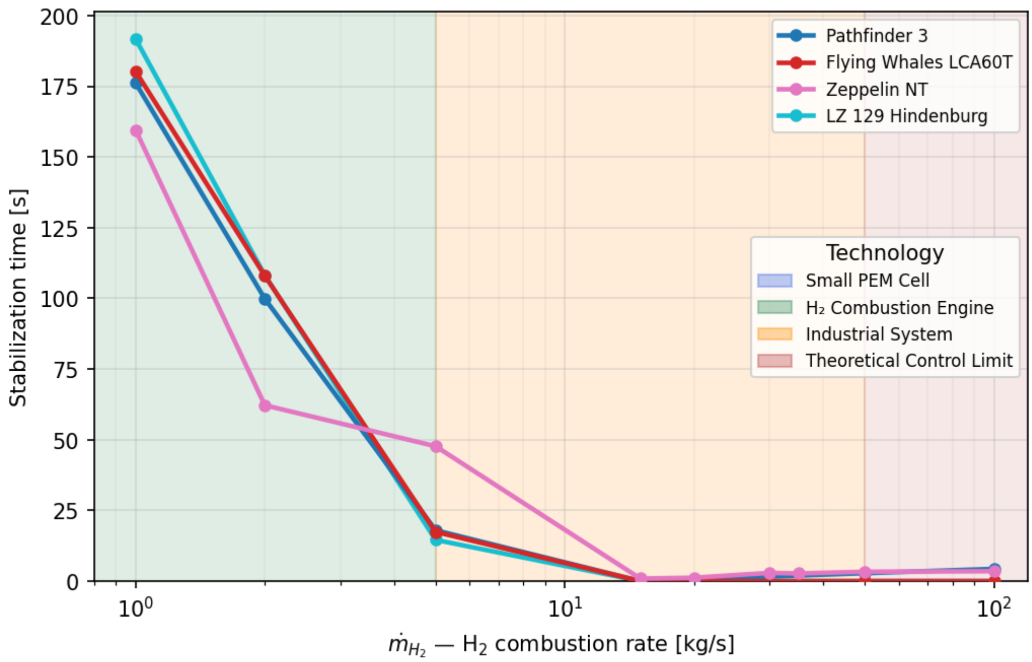

Figure 5. Parametric sweep: stabilization time t_{stab} as a function of the H_2 combustion rate $\dot{m}_{H_2}$ for each evaluated airship, accounting for O_2 mass absorption. t_{stab} is reached when $|\Delta h| < 1$ m and the residual vertical velocity falls below 0.05 m/s. Technology bands follow Figure 4. At low rates (combustion-engine range), stabilization times exceed 100 s for all platforms. The Zeppelin NT exhibits faster stabilization relative to large-scale platforms within the industrial-system range, consistent with its lower inertia; its $\dot{m}_{H_2,\text{min}}$ crossing at 20 kg/s corresponds to a stabilization time of ≈48 s, reflecting the marginal nature of the safety-threshold crossing at minimum viable flow. All platforms converge below 5 s at rates exceeding 50 kg/s.

Figures 4 and 5 confirm the trends from Eq. (11). The **Pathfinder 3** satisfies the safety threshold at 7 kg/s and the **Flying Whales LCA60T** at 6 kg/s, consistent with the analytical prediction of 6.6 and 5.8 kg/s respectively (within 6% and 3%). The **LZ 129 Hindenburg** crosses the threshold at 4 kg/s (analytical: 4.1 kg/s, 2% deviation). The **Zeppelin NT** crosses the 5 m threshold at 20 kg/s and the precision threshold at 35 kg/s; its simulated minimum falls 25% below the analytical purely-reactive bound (26.6 kg/s), consistent with the greater relative benefit of proactive preconditioning at $\varepsilon = 0.140$. It is notable that at extreme rates ($\geq$ 100 kg/s), the displacement $|\Delta h|$ becomes slightly negative (e.g., -0.023 m for Pathfinder 3). This is not a numerical artifact, but evidence that the -10 kg/s preconditioning, not being limited by saturation of the main actuator in this phase, reduces buoyancy so aggressively that the airship initiates a controlled descent before the payload loss can induce any ascent. From Figure 5, the Zeppelin NT achieves the shortest stabilization time (3.2 s at $\dot{m}_{H_2,\text{opt}} = 35$ kg/s) despite its highest flow requirement, while the Hindenburg requires the longest (12.4 s), consistent with its larger inertia slowing both the perturbation and the recovery dynamics.

3.1.2. Minimum and Optimal Rates

From the discrete parametric sweep, two operational thresholds are defined to classify the extraction rates $\dot{m}_{H_2,max}$:

- $\dot{m}_{H_2,\text{min}}$: the minimum value in the evaluated set that guarantees $|\Delta h| \leq 5$ m, in accordance with external-load operational criteria [13,14] (see Section 2.2).
- $\dot{m}_{H_2,\text{opt}}$: the minimum value in the evaluated set that achieves $|\Delta h| \leq 1$ m, the precision threshold for structural coupling operations (see Section 2.2).

Table 3. Minimum and optimal H_2 combustion rates for a 1000 kg payload release. Values correspond to the discrete bounds of the parametric sweep.

Airship	**Safety** ($\lvert\Delta h\rvert \leq 5$ m)		**Precision** ($\lvert\Delta h\rvert \leq 1$ m)			
	$\dot{m}_{H_2,\mathrm{min}}$ (kg/s)	$\lvert\Delta h\rvert$ (m)	$\dot{m}_{H_2,\mathrm{opt}}$ (kg/s)	$\lvert\Delta h\rvert$ (m)	t_{stab} (s) (at $\dot{m}_{H_2,\mathrm{opt}}$)	Analytical $\dot{m}_{H_2,\mathrm{min}}$ (kg/s) (Eq. (11))
LZ 129 Hindenburg	4	4.697	8	0.988	12.4	4.1
Pathfinder 3	7	3.286	20	0.339	4.4	6.6
Flying Whales	6	3.901	10	0.958	10.2	5.8
Zeppelin NT	20	3.359	35	0.650	3.2	26.6[a]

[a] Reactive analytical bound; simulated value 25% lower due to preconditioning benefit at $\varepsilon = 0.140$.

From a technological standpoint, the requirements found dictate the type of subsystem needed. For the Hindenburg ($\dot{m}_{H_2,\mathrm{min}} = 4$ kg/s) and Flying Whales LCA60T ($\dot{m}_{H_2,\mathrm{min}} = 6$ kg/s), the requirement falls within the range of industrial H_2 combustion engines or large-scale premix burners. The Pathfinder 3 ($\dot{m}_{H_2,\mathrm{min}} = 7$ kg/s) demands similar hardware at slightly higher throughput. The Zeppelin NT demands the highest flow ($\dot{m}_{H_2,\mathrm{min}} = 20$ kg/s), requiring aeroderivative gas turbines adapted for hydrogen or high-pressure injection systems. The peak post-release acceleration for each platform is bounded by $\varepsilon\, g$ and remains below $0.2\, g$ across all evaluated cases (Section 4.3).

3.1.3. Dynamic Validation and Optimization Comparison

Two controller configurations are evaluated on the Pathfinder 3 platform, corresponding to the two analytical frameworks of Sections 2.5 and 2.6: a *standard scheme* with hard-coded preconditioning parameters ($\dot{m}_{\mathrm{prec}}$=-10 kg/s, t_{anticip}=2 s, $\dot{m}_{H_2,\mathrm{max}}$=30 kg/s) and a *minimax-optimal scheme* in which all parameters are derived analytically from Appendix A (α^*=0.215, $\dot{m}_{\mathrm{prec}}$=-5.21 kg/s, t_{anticip}=3.83 s, $\dot{m}_{H_2,\mathrm{max}}$=$\dot{m}^*_{H_2}$=5.21 kg/s).

Standard scheme.

Figure 6 shows the single-release base case on a logarithmic time axis, which expands the resolution of the critical instants around the release at $t = 1.0$ s. The maximum vertical displacement is contained at $\Delta h \approx 0.37$ m, within the precision threshold (1 m). The proactive H_2 extraction pulse is clearly visible in the flow panel prior to release; peak acceleration remains below $0.1\, g$ throughout. A key physical mechanism is the O_2 mass absorption from the atmosphere: for every kilogram of H_2 burned, the system ingests 8 kg of atmospheric oxygen producing 9 kg of liquid water that acts as inertial ballast — captured by the L_{eff} factor in Eq. (12). Generated energy: ≈460 kWh per delivery cycle. A 4-drop stress test ($\Delta t = 30$ s) further confirms robustness: the adaptive gain $K_d(t)$, recalculated at each drop to maintain $\zeta = 1.35$, prevents oscillation amplification despite the step-wise reduction in inertial mass, with cumulative generated energy ≈1,785 kWh.

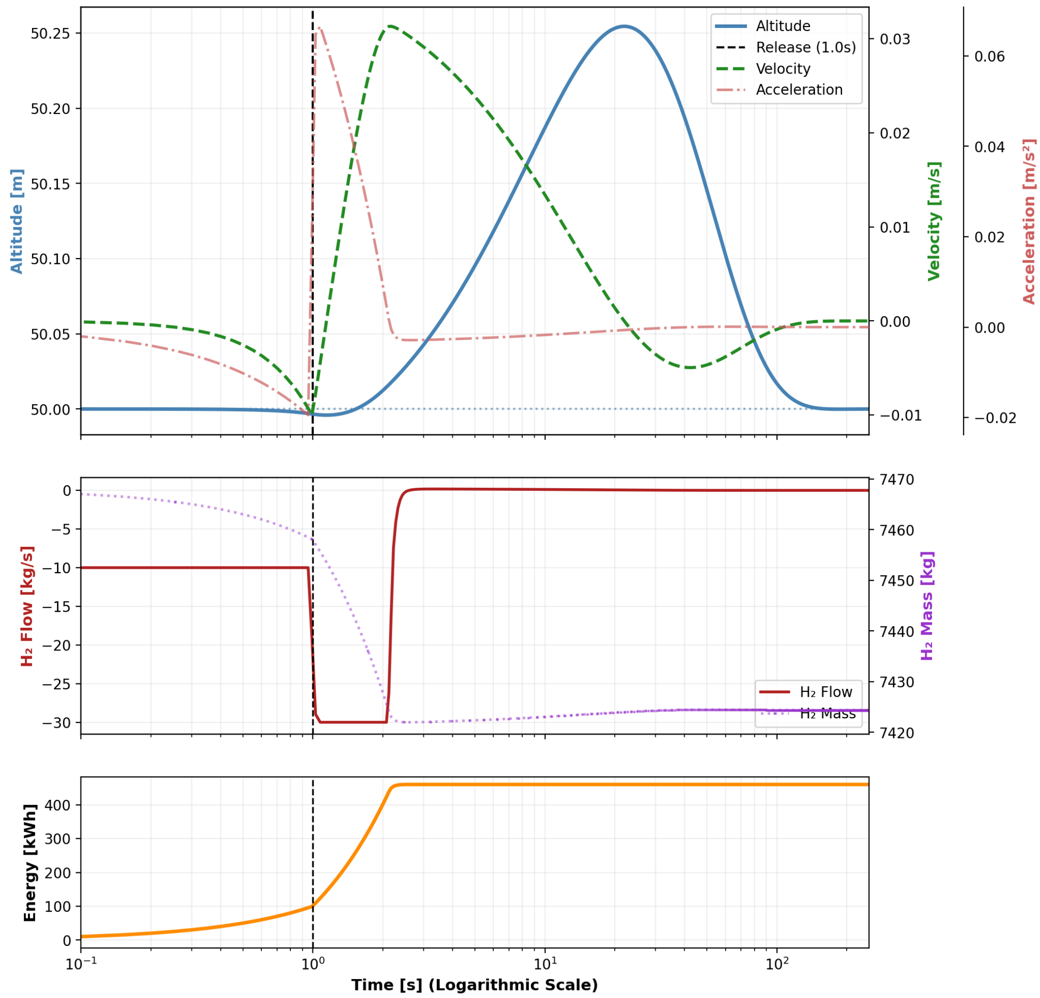

Figure 6. Standard scheme — Base Mission: dynamic simulation of the Pathfinder 3 ($\dot{m}_{H_2,\max}$=30 kg/s, $\dot{m}_{\text{prec}}$=$-$10 kg/s, t_{anticip}=2 s) for a single 1-tonne release. Logarithmic time axis resolves the preconditioning pulse and post-release transient around t=1 s. Peak displacement $\Delta h \approx 0.37$ m satisfies the precision threshold (1 m).

Minimax-optimal scheme.

Applying the minimax formulation of Section 2.6 to the Pathfinder 3, the system constant $C = 0.127$ places it in the high-inertia regime ($\Delta > 0$), yielding $\alpha^* = 0.215$ via Cardano's formula. The preconditioning parameters are recalculated analytically as $\dot{m}_{\text{prec}} = -5.21$ kg/s and $t_{\text{anticip}} = 3.83$ s, reducing the required installed actuator capacity to $\dot{m}^*_{H_2} = 5.21$ kg/s — an 83% reduction relative to the standard configuration.

Figure 7 shows the base case with the minimax-optimal parameters on a linear time axis, resolving the extended anticipation window and the post-release recovery arc. Peak displacement reaches $\Delta h \approx 4.2$ m, within the safety threshold (5 m). The system converges to $(h, v) \to (h_0, 0)$ through the combined action of the overdamped PD controller ($\zeta = 1.35$) and aerodynamic drag, confirming that the minimax formulation provides the minimum installed capacity such that full state convergence is achievable — force balance is guaranteed analytically; velocity convergence is ensured by the closed-loop PD dynamics. Generated energy: 964 kWh, consistent with Eq. (17) within the aerodynamic drag margin.

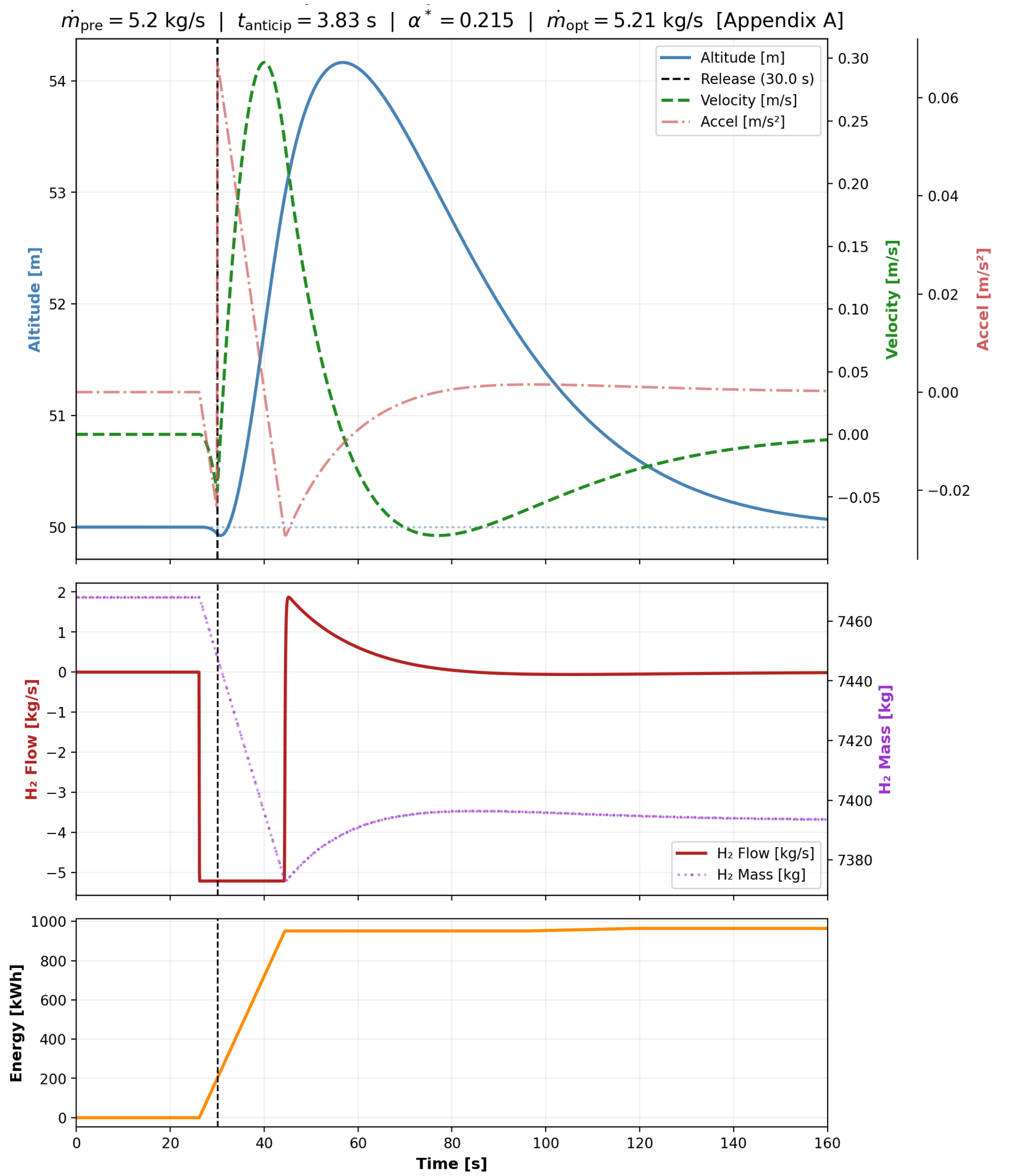

Figure 7. Minimax-optimal scheme — Base Mission: dynamic simulation of the Pathfinder 3 with analytically derived parameters ($\dot{m}^*_{H_2}$=5.21 kg/s, $\dot{m}_{prec}$=−5.21 kg/s, $t_{anticip}$=3.83 s, α^*=0.215; Appendix A). Linear time axis. Peak displacement $\Delta h \approx 4.2$ m satisfies the safety threshold (5 m). Generated energy: 964 kWh.

Figure 8 presents the 4-drop stress test under the optimal scheme. Both $\dot{m}_{prec}$ and $t_{anticip}$ are recalculated analytically before each drop to compensate for the decreasing vehicle mass ($\dot{m}_{prec}$ from 5.14 kg/s at drop 1 to 5.21 kg/s at drop 4; $t_{anticip}$ from 3.89 s to 3.84 s), maintaining α^*=0.215 across the full unloading sequence. All four displacements remain within the safety threshold; cumulative generated energy: 3,586 kWh (≈897 kWh/drop, consistent with the single-drop result within 7%).

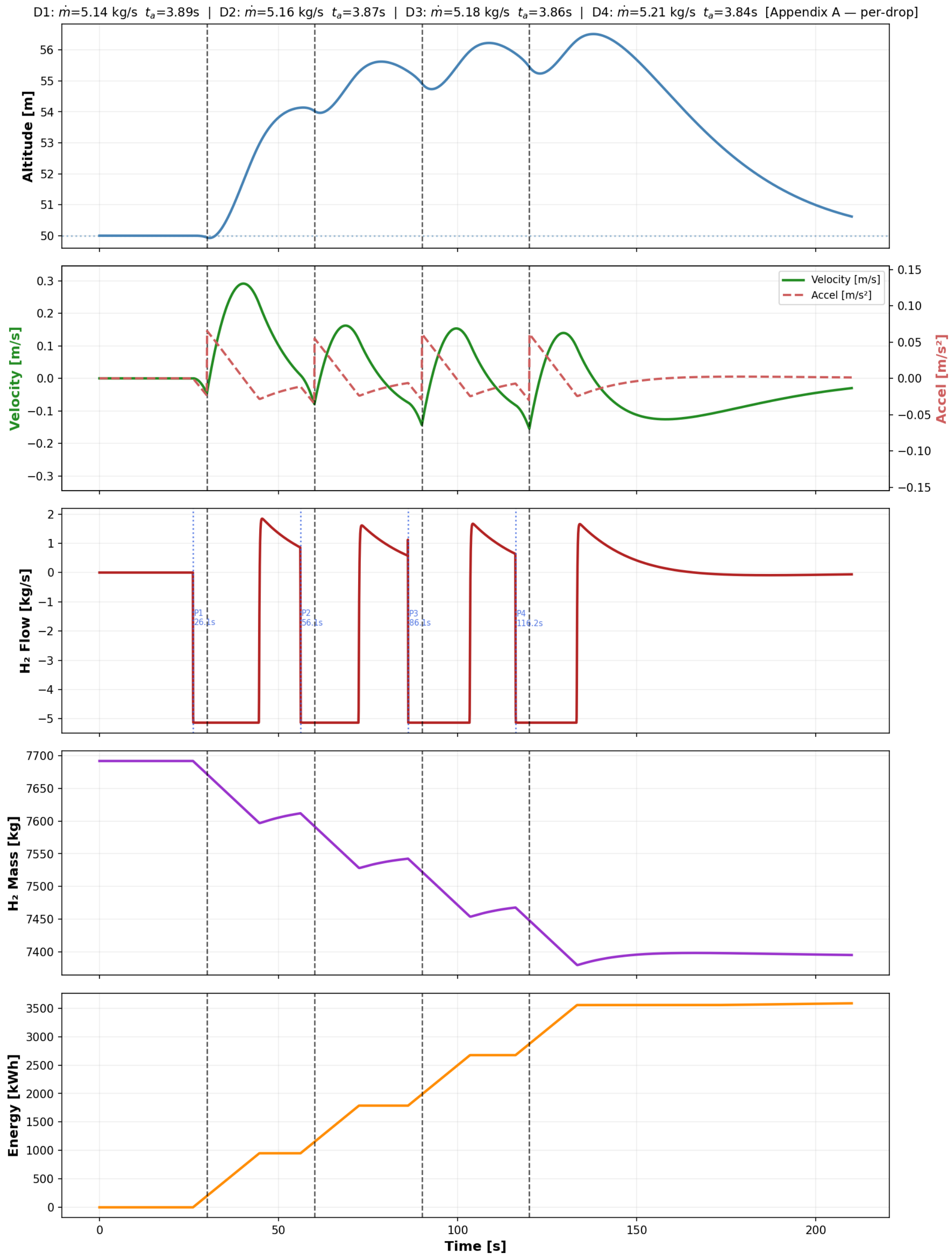

Figure 8. Minimax-optimal scheme — Stress Test: multi-drop mission of the Pathfinder 3 ($\dot{m}^*_{H_2}$=5.21 kg/s) with 4 consecutive 1-tonne releases. Optimal flow $\dot{m}_{\text{prec}}$ (from 5.14 to 5.21 kg/s) and anticipation time t_{anticip} (from 3.89 s to 3.84 s) recalculated analytically before each drop to compensate for step-wise mass reduction. Cumulative generated energy: 3,586 kWh.

Comparative summary.

The two configurations differ primarily in the installed actuator capacity and the resulting operating margin. The standard scheme ($\dot{m}_{H_2,\text{max}} = 30$ kg/s, $\dot{m}_{\text{prec}} = -10$ kg/s, $t_{\text{anticip}} = 2.0$ s) operates with a $6\times$ oversized actuator relative to the analytical minimum, achieving a peak displacement of $\Delta h_{\max} \approx 0.37$ m well within the precision threshold (≤ 1 m) and recovering ≈460 kWh per delivery. The minimax-optimal scheme ($\dot{m}^*_{H_2} = 5.21$ kg/s, $\dot{m}_{\text{prec}} = -5.21$ kg/s, $t_{\text{anticip}} = 3.83$ s, $\alpha^* = 0.215$) reduces the installed hardware by 83% at the cost of operating closer to the safety boundary ($\Delta h_{\max} \approx 4.2$ m within the 5 m threshold), with a higher per-cycle electrolysis demand (≈964 kWh) recoverable through the photovoltaic regeneration cycle. In both configurations, the adaptive PD controller and the O_2 mass-absorption mechanism ensure full state convergence and energetic closure of the regenerative cycle.

Multi-platform validation.

The minimax-optimal scheme is extended to the remaining three platforms (Hindenburg, Flying Whales, Zeppelin NT), using the per-drop analytical recalculation framework of Appendix A. Table 4 consolidates the resulting parameters for the single 1-tonne release; the universality of $\alpha^* = 0.215$ across all four platforms confirms the platform-independence of the optimal preconditioning fraction. The required actuator capacity $\dot{m}^*_{H_2}$ scales as $M_{\text{total}}^{-1/2}$, consistent with the scaling paradox predicted analytically in Section 2.6: larger airships require lower mass flows for equivalent payload perturbations. Generated energy per cycle (958–985 kWh) is approximately independent of platform inertia, since it depends only on the H_2 mass burned per kg of cargo released — the energetic signature of the regenerative cycle is platform-invariant.

Table 4. Minimax-optimal scheme: per-platform analytical parameters for a single 1-tonne release. $\alpha^* = 0.215$ is universal; $\dot{m}^*_{H_2} \propto M_{\text{total}}^{-1/2}$ confirms the scaling paradox.

Platform	M_{total} (kg)	α^*	$\dot{m}^*_{H_2}$ (kg/s)	t_{anticip} (s)	E_{gen} (kWh)
LZ 129 Hindenburg	284,360	0.2150	3.20	6.24	985
Flying Whales LCA60T	139,708	0.2150	4.57	4.37	966
Pathfinder 3	107,468	0.2150	5.21	3.83	964
Zeppelin NT	7,117	0.2150	20.25	0.99	958

The Pathfinder 3 simulations (Figures 7 and 8) provide a visual baseline of the minimax-optimal dynamics; analogous time-history plots were generated for the three additional platforms following the same minimax-optimal scheme. For the Hindenburg and Flying Whales platforms, the multi-drop stress test consists of four consecutive 1-tonne releases. For the Zeppelin NT, the operationally representative configuration is four consecutive 250 kg releases: a single 1-tonne drop represents 14% of the platform's total mass (the highest ε of the evaluated set, Table 2) and pushes the actuator into the $\dot{m}^*_{H_2} = 20.25$ kg/s extreme of Table 4, retained as the scaling-paradox reference; the 4×250 kg sequence aggregates to the same total cargo while keeping each individual perturbation within typical operational regimes. In every case, $\alpha^* = 0.215$ holds invariant, the analytical per-drop recalculation of $\dot{m}^*_{H_2}$ and t_{anticip} tracks the step-wise mass reduction within $\leq 1\%$ between consecutive drops, and the cumulative generated energy scales linearly with the number of releases: 3,222 kWh (Hindenburg multi), 3,493 kWh (Flying Whales multi), and 1,123 kWh (Zeppelin NT multi 4×250 kg).

The simulations confirm that the minimax-optimal scheme is robust across the full $40\times$ range of structural masses considered (Zeppelin NT to Hindenburg), with $\alpha^* = 0.215$ remaining invariant and the required actuator flow following the analytical scaling law $\dot{m}^*_{H_2} \propto M_{\text{total}}^{-1/2}$.

3.2. Energy Balance and Auxiliary Subsystems

3.2.1. Condensation and General Balance

With atmospheric O_2 as the oxidizer, the effective specific lift factor L_{eff} (Eq. (12)) governs both the preconditioning dynamics and the descent-phase H_2 consumption. Each kilogram of H_2 burned creates $L_{\text{eff}}\,g$ of net downward force through the combined effect of buoyancy loss and atmospheric O_2 mass absorption, so the H_2 required during the descent stabilization phase is:

$$m_{H_2,\text{desc}} = \frac{m_{\text{load}}}{L_{\text{eff}}} = \frac{m_{\text{load}}}{\rho_{\text{air}}/\rho_{H_2} + 8} \approx 44.7\,\text{kg} \tag{15}$$

This produces $m_{H_2O,\text{desc}} = 9 \times 44.7 \approx 402$ kg of water vapor in the critical condensation window. The time to recover it as liquid ballast is:

$$t_{\text{cond}} = \frac{m_{H_2O,\text{desc}} \cdot L_v}{Q_{\text{cond}}}, \quad L_v = 2257\,\text{kJ/kg} \tag{16}$$

From Eq. (16), setting $t_{\text{max}} = 10$ min as the operational cadence limit, the minimum condenser power is $Q_{\text{cond,min}} \approx 1{,}512$ kW (Fig. 9). The descent-phase stabilization acts as a secondary energy source. With $\eta_{\text{gen}} = 0.30$ [18]:

$$E_{\text{gen,desc}} = m_{H_2,\text{desc}} \cdot \text{PCI}_{H_2} \cdot \eta_{\text{gen}} \approx 447\,\text{kWh} \tag{17}$$

This descent-phase energy recovery, ≈447 kWh per delivery, is consistent with the dynamic validation results of Section 3.1.3 (460–964 kWh; the range reflects differing preconditioning depths and actuator capacities across the two validated configurations).

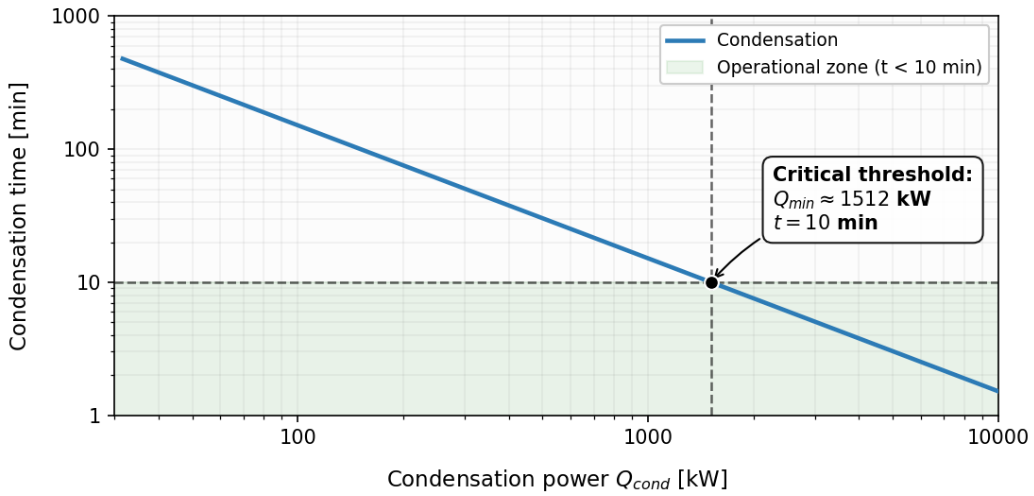

Figure 9. Condensation time vs. Q_{cond} for the descent-phase water output ($m_{H_2O,\text{desc}} \approx 402$ kg, O_2-corrected). The $Q_{\text{cond,min}} \approx 1{,}512$ kW threshold corresponds to a 10 min recovery window for the descent-phase water load.

This energy surplus constitutes a direct input for the electrolysis bank, optimizing the overall fuel recovery balance detailed in the following section.

4. Discussion

4.1. Solar Regeneration and Operational Autonomy

Solar energy management and onboard hydrogen regeneration have been studied as autonomy-extending mechanisms in stratospheric airship platforms [11,26]. Building on this approach, the capacity to recover consumed hydrogen is modeled as a function of the available daily irradiance and the useful surface area of the vehicle. Daily regeneration is calculated using:

$$\dot{m}_{H_2,\text{reg}} = \frac{r_{\text{ef}} \cdot f_{\text{pan}} \cdot A_{\text{total}} \cdot \eta_{\text{el}}}{E_{H_2,\text{el}}} \tag{18}$$

adopting as design parameters $f_{\text{pan}} = 0.45$ (maximum effective area assuming thin-film solar cells on the upper curvature), $\eta_{\text{el}} = 0.65$ and an electrolysis energy demand of $E_{H_2,\text{el}} = 45$ kWh/kg, the lower bound of the 45–60 kWh/kg range reported for state-of-the-art PEM electrolysis [33,47].

Under average irradiance conditions in Lima (2.0 kWh/m²/day, equivalent to 2.0 peak sun-hours/day in solar-industry standard notation), vehicle scale dictates the viability of the closed cycle. For a single 1-tonne release, the hydrogen demand per delivery is ≈44.7 kg (descent phase, O_2-corrected; see Section 3.2), which sets the electrolysis regeneration target. The **LZ 129 Hindenburg** (0.14 days, 3.4 h) and the **Flying Whales** (0.15 days, 3.6 h) stand out as the efficiency benchmarks thanks to their large surface areas (24,000 and 23,000 m^2 respectively). By contrast, the **Pathfinder 3** requires 0.26 days (6.4 h), while the **Zeppelin NT** (36 kg/day) is penalized with a recovery cycle of approximately 29.5 hours (1.23 days), exceeding the standard 24-hour delivery cadence.

Scale effects become decisive in high-intensity mission profiles. Faced with a demand of 4 consecutive releases (∼179 kg of H_2), both the **Hindenburg** (0.57 days) and the **Flying Whales** (0.60 days) maintain daily self-sufficiency. The **Pathfinder 3** narrowly exceeds its daily cadence at 1.06 days, while the **Zeppelin NT** (≈4.9 days) would greatly exceed its immediate solar conversion rate. In these cases, the limitation does not lie in flight autonomy but in the water-to-gas conversion speed; the vehicle will need to draw from its pressurized H_2 reserve tanks while the system progressively regenerates the stored liquid ballast. This contrast underscores that the regenerative cycle is naturally optimized in large-scale airships, where the high area-to-mass ratio favors continuous unloading logistics without depleting main reserves or depending on external supply.

4.2. Comparison with Existing Buoyancy Control Approaches

Table 5 positions the proposed cycle within the landscape of existing buoyancy control approaches, evaluated along three operational axes: per-cycle material loss, cycle closure, and flight demonstration status.

Table 5. Comparison of post-release buoyancy compensation approaches.

Approach	**Material loss**	**Closed-cycle**	**Dynamic stab.**[c]	**Flight demo**	**Ref.**
Passive gas venting	Yes (H_2)	No	Yes	Yes	[2]
Consumable ballast	Yes (ballast)	No	Yes	Yes	[2]
Sliding ballast	No	Partial[a]	Partial[a]	Yes	[7]
Inflatable vacuum chambers	No	Yes	No	No	[8]
Active gas transfer	Yes (H_2)	No	Partial	Partial	[9]
He compression (COSH)	No	Yes[b]	No[b]	Yes (prototype)	[10]
H_2 regen. fuel cell	No	Yes	No	No	[11]
This work	**No**[d]	**Yes**	**Yes**	**No (sim.)**	—

[a] Provides attitude control but does not address net buoyancy surplus after release.
[b] Closed-cycle via mechanical He compression; does not integrate with the H_2 energy chain; no post-release stabilization.
[c] Active compensation of the post-release aerostatic imbalance within the safety threshold ($|\Delta h| \leq 5$ m).
[d] Under ideal closed-cycle assumption.

Among closed-cycle approaches, COSH relies on mechanical He compression within a separate gas infrastructure and provides no post-release dynamic stabilization; regenerative fuel cells address pressure maintenance but not the transient aerostatic imbalance following a payload release. The proposed cycle is the only approach in Table 5 that simultaneously achieves closed-cycle operation, no per-cycle material loss (under ideal closed-cycle assumption; Table 5, note *d*), and active post-release stabilization within a single H_2 chain. Beyond the system concept, this work provides analytical tools absent from prior approaches: a closed-form scaling law for the minimum actuator capacity (Eq. (11)), a minimax formulation with exact solution via Cardano's method (Section 2.6, Appendix A), and a parametric characterization validated across four representative platforms.

4.3. Scale and Sensitivity

The most notable result is the inversion of scale intuition: the Hindenburg (264 t) requires the lowest rate (4 kg/s), while the Zeppelin NT (5.6 t) is the most sensitive (20 kg/s). This paradox has a direct physical explanation: a higher total mass implies a smaller perturbation index $\varepsilon = m_{\text{load}}/M_{\text{total}}$, which in turn yields a longer available response window τ_{ctrl} before the safety threshold is exceeded (Eq. (10)). The actuator therefore operates against a slower-evolving perturbation, reducing the required instantaneous flow rate. This result positions the regenerative H_2 cycle as intrinsically suited for next-generation heavy-lift platforms, where it is precisely the largest vehicles that benefit most.

Beyond the flow rate requirement, ε also governs the peak post-release acceleration: $a_{\max} = \varepsilon\, g$, bounded by $0.140\, g$ for the Zeppelin NT — the most dynamically sensitive platform — and below $0.01\, g$ for the three large-scale cases. All evaluated platforms therefore satisfy an operational comfort threshold of $0.2\, g$ without additional constraint, confirming that the perturbation is dynamically significant for buoyancy control but does not impose structural or personnel-safety hazards in any of the evaluated cases.

4.4. Stabilization Scope: Force Balance, State Convergence, and Fine-Attitude Control

The minimax formulation of Section 2.6 determines the minimum installed capacity $\dot{m}^*_{H_2}$ such that the aerostatic force balance is achievable within the preconditioning depth D_{lim} — a necessary condition for stabilization. Full state convergence, $(h, v) \to (h_0, 0)$, is a stronger requirement that the minimax alone does not guarantee analytically: it is ensured by the overdamped PD controller ($\zeta = 1.35$) acting within the capacity provided by $\dot{m}^*_{H_2}$, as confirmed numerically in Section 3.1.3. Once force balance is achievable, the derivative term $K_d\, v$ continuously injects corrective flow proportional to the residual velocity, driving the state to rest; aerodynamic drag provides additional passive damping that reinforces convergence monotonically.

For operations at the absolute minimum installed capacity — where the actuator is fully saturated throughout the maneuver — a small residual velocity may persist after the buoyancy system has exhausted its authority. In practice, this residual is handled by the fine-attitude propulsion system (vectored thrusters or differential propeller pitch) employed during standard airship approach and landing procedures [13,14]. This two-layer architecture — buoyancy system for gross perturbation attenuation, fine-attitude propulsion for terminal velocity correction — imposes no additional hardware requirements beyond the existing propulsion system and is consistent with established operational airship protocols.

4.5. Condensation, O_2 Source and Mass Balance

The threshold $Q_{\text{cond,min}} = 1{,}512$ kW corresponds to the scale of medium-capacity industrial heat exchangers used in chemical processing applications. The analysis adopts atmospheric O_2 as the reference choice, although the system equally admits onboard tanks. As demonstrated quantitatively in Figure 10, atmospheric air reduces the descent-phase H_2 burning requirement by ≈35.8% relative to onboard O_2 (from ≈69.6 kg to 44.7 kg per tonne). The physical mechanism is as follows: with onboard O_2, only the buoyancy reduction contributes to stabilization (effective factor $\rho_{air}/\rho_{H_2} \approx 14.4$); with atmospheric air, the stoichiometric 8 kg of O_2 absorbed per kg of H_2 adds directly to the vehicle inertial mass, yielding $L_{eff} = \rho_{air}/\rho_{H_2} + 8 \approx 22.4$, consistent with the O_2-corrected formula of Section 3.2. The required air ingestion (≈34 kg of air per kg H_2 at stoichiometry) is handled naturally by the modular combustor architecture: each 1 kg/s module operates with ≈34–68 kg/s of combustion air at $\lambda = 2$–3, consistent with the GE90-class core airflow demonstrated in [31] (≈130 kg/s per combustion chamber). This supplementary inertial ballast simultaneously reduces actuator flow demand and the critical-window condensation load, constituting an additional degree of freedom for mission profile design.

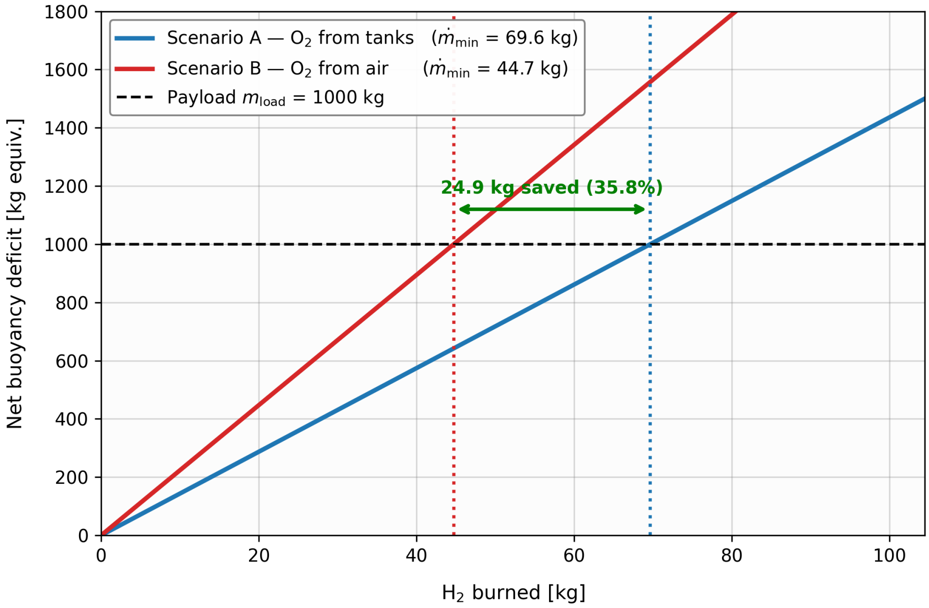

Figure 10. Comparative analysis of the O_2 source (onboard tanks vs. atmospheric air) for the regenerative cycle after the release of 1 tonne of payload. Using atmospheric air reduces the descent-phase H_2 demand by ≈35.8% (from ≈69.6 kg to ≈44.7 kg; Eq. (15)).

4.6. Hardware Implementation Constraints

The actuator-sizing laws derived in Section 2.6 fix the parametric ranges over which each subsystem of the regenerative cycle must operate. Table 6 details the alignment between these model-derived requirements and the operational ranges demonstrated in peer-reviewed aerospace and industrial hardware; a detailed subsystem-by-subsystem analysis with mass bounds is provided in Appendix C.

Table 6. System-level requirements derived from the actuator-sizing laws of Section 2.6, contrasted with demonstrated aerospace and industrial benchmarks. Full subsystem mass bounds: Appendix C.

Subsystem	Required by model	Demonstrated benchmark
Generator unit	$\dot{m}^*_{H_2} = 2.5$–5.2 kg/s; 300–624 MW (burst-duty ∼30 s/cycle)	1.0–1.25 kg/s per GE90-class H_2 combustor, 121–150 MW thermal (CFD) [31]
Condenser	$Q_{\mathrm{cond}} \geq 1{,}512$ kW	HySIITE aerospace rig: 1.26 kg/s H_2O, ∼2,844 kW; 20–50 kW/kg density [27]
PEM electrolyser	84 kW continuous; 44.7 kg H_2/day; $\eta \geq 65\%$; ≥60,000 h	0.5–2.2 kW/kg [30,41]; ≥60,000 h [47]
Photovoltaic array	Close daily H_2 regeneration budget (84 kW electrolysis, 44.7 kg/day)	$f_{\mathrm{pan}} = 0.45 \cdot A_{\mathrm{total}}$ of airship-grade thin-film (0.12 kg/m^2, $\eta = 8\%$) closes budget under mean irradiance [32]
H_2 storage (COPV)	30–105 kg H_2/cycle at 350 bar	Type-IV composite, 5.5 wt% gravimetric density; SAE J2601-2 [17]
H_2 compressor	∼6 kW at ∼1.86 kg/h; 30–350 bar	Hofer Case 2: 11 kW at 1.8 kg/h, 2-stage diaphragm [33,45,46]
Envelope & ballonets	$\Delta P = 1$–50 mbar; multilayer laminate	5 mbar (Zeppelin NT) [36]; Tedlar/polyester/polyurethane [48]

The integrated mass budget is detailed in Appendix C.8 and summarised in Table 7. The optimistic component bound applies advanced HT-PEM at 50 kg [30] and burst-duty generator modules at 250 kg each [37]; the pessimistic bound applies air-cooled PEM at 170 kg and continuous-duty generator modules at 600 kg each. The viability column reports ✓ when the system mass remains within the platform's payload capacity and × in the scaling-limit regime where it does not.

Table 7. Per-platform mass envelope. Each platform reports two rows: optimistic component bound (upper) and pessimistic component bound (lower); see preceding paragraph for the bound definitions.

Platform	Payload (kg)	System mass (kg)	% payload	Viability
LZ 129 Hindenburg	12,500	4,309	34%	✓
		5,829	47%	✓
Flying Whales LCA60T	60,000	4,644	7.7%	✓
		6,514	10.9%	✓
Pathfinder 3	10,000	3,049	30%	✓
		5,269	53%	✓
Zeppelin NT (4×250 kg)	1,000	1,831	183%	×
		3,001	300%	×

Three of four platforms satisfy the mass envelope under both optimistic and pessimistic component bounds, with mass-budget overhead between 7.7% and 53% of payload. The Zeppelin NT exceeds the payload threshold under any current technology assumption, consistent with the scaling paradox of Section 2.6.

The thin-film amorphous baseline (120 g/m^2, $\eta = 8\%$ [32]) brings three of four platforms to sub-24 h regeneration; the Zeppelin NT remains borderline at 29.5 h (Section 3.2). Higher-efficiency airship-grade alternatives (single-junction $\eta = 24\%$, triple-junction $\eta = 32\%$) shorten the regeneration cycle in inverse proportion to η at a proportionate mass penalty; the per-platform impact is detailed in Appendix C.3.

The energy balance assumes complete water recovery ($\eta_{\mathrm{cond}} = 1$). Partial condensation ($\eta_{\mathrm{cond}} \approx 0.85$) does not degrade buoyancy control — the feedback PD compensator adapts to the residual disturbance — but proportionally increases the H_2 flow needed to close the cycle, absorbed by the photovoltaic regeneration loop; a unified electrolyser–fuel-cell architecture (Appendix C.7) would consolidate this subsystem further.

Combustion at the nominal 4–20 kg/s range releases 480 MW–2.4 GW of thermal power at atmospheric pressure. Lean premix operation ($\lambda > 2$) keeps the adiabatic flame temperature below 900 °C [29,34] — compatible with commercial high-temperature alloys and the equipment requirements of Directive ATEX 2014/34/EU — and the ballonet material is isolated from the combustion zone by an upstream pre-chamber, as in standard industrial H_2 boilers. The wide flammability range of H_2 in air (4–75% by volume [28]) is bounded by the lean premix condition, which simultaneously suppresses NO_x formation [34].

The model adopts a conservative generator efficiency $\eta_{\text{gen}} = 0.30$; modern H_2 combustion systems reach $\eta_{\text{gen}} \approx 0.40$–$0.55$ [18,42–44], which would not change the H_2 consumption —which depends only on the aerostatic balance (Eq. (15))— but would scale the electrical energy recovered per cycle linearly through Eq. (17). Added-mass effects [9] are neglected, consistent with the low accelerations of the controlled stabilization regime. The minimax-optimal scheme entails a hardware–fuel trade-off documented in Section 3.1.3: lower installed actuator capacity at the cost of higher per-cycle electrolysis demand.

4.7. Future Work

Methodological extensions include multi-axis attitude control, formal integration with the fine-attitude propulsion system (Section 4.4), and replacement of the hybrid PD scheme by a model predictive controller (MPC) operating on D_{lim} and α^*. On the engineering side, system-level integration and certification under aerospace airworthiness frameworks remain the primary challenges. Experimental subsystem-level validation and the definition of the *ideal airship* — the platform that simultaneously minimises all actuator, condenser, and regeneration requirements — constitute primary directions for this research area.

5. Conclusions

1. The **dynamically validated** optimal flow rates (Section 3.1.3, Table 4) span 3.2 kg/s (LZ 129 Hindenburg) to 20.25 kg/s (Zeppelin NT, single 1-tonne extreme); the operationally representative configuration of the Zeppelin NT (multi-drop 4×250 kg) settles at $\sim$2.5 kg/s. These rates fall within ranges currently demonstrated by aerospace H_2 combustion technology (Table 6), and the perturbation index $\varepsilon = m_{\text{load}}/M_{\text{total}}$ orders the requirements monotonically: the higher the ε, the higher the rate.
2. The **anticipatory preconditioning** reduces $\dot{m}_{H_2,\text{min}}$ below the purely-reactive analytical bound ($\approx$27 kg/s for the Zeppelin NT, $\tau_{\text{ctrl}} \approx 2.6$ s), and the closed-form **minimax formulation** (Appendix A) further compresses the installed capacity to $\dot{m}^*_{H_2} \in [3.2, 20.25]$ kg/s across the four platforms, with the compensation fraction $\alpha^* = 0.215$ holding invariant over the full $40\times$ range of structural masses — a universal design constant emerging directly from the closed-form analytical structure. For the Pathfinder 3 representative case, this corresponds to an 83% reduction relative to the 30 kg/s standard configuration, with numerical validation within 15–25% of the simulated values.
3. The use of **atmospheric O_2 as oxidizer** cuts the descent-phase H_2 requirement by $\approx$36% relative to an onboard-O_2 scheme ($\approx$44.7 vs. $\approx$70 kg per tonne released). The absorbed O_2 mass simultaneously acts as supplementary inertial ballast, lowering the actuator demand and relaxing the critical-window condensation requirement, while opening an additional degree of freedom in mission profile design.

4. **Larger airships are easier to stabilize**. All four evaluated airships satisfy the safety threshold ($|\Delta h| \leq 5$ m) with achievable rates; large-scale models operate the deepest within the threshold and with the smallest specific actuator load.
5. **Three of the four evaluated airships** achieve energetic self-sufficiency through solar regeneration under average Lima irradiance, recovering the H_2 consumed per delivery in under one day. The Zeppelin NT is the exception: its limited upper-surface coverage (1,260 m^2 of usable photovoltaic area, against 10,800 m^2 for the LZ 129 Hindenburg) extends the recovery cycle to approximately 30 hours (1.23 days), beyond the 24-hour delivery cadence — the regenerative cycle is naturally optimised for large-scale platforms.
6. The regenerative cycle produces a **quantifiable and predictable hydrogen demand** at a single scale — ≈44.7 kg per tonne of payload (descent phase, O_2-corrected) — governing condenser sizing, actuator requirements, and solar regeneration planning alike. This predictability feeds directly into techno-economic feasibility studies of H_2 supply chains for aerial heavy-lift logistics.
7. The parametric analysis yields a complete set of subsystem specifications consolidated in Tables 6 and 7: a condenser of $Q_{\text{cond,min}} \geq 1{,}512$ kW for the 10-min critical window; a generator unit in burst-duty operation at $\dot{m}^*_{H_2} = 2.5$–5.2 kg/s, 300–624 MW thermal (Table 6; per-module sizing: Appendix C.2); a PEM electrolyser of ≈84 kW continuous capacity sustaining 44.7 kg H_2/day; a Type-IV COPV storage system of 30–100 kg H_2/cycle at 350 bar; and a photovoltaic array covering ≥45% of the upper-surface area. Each subsystem operates within ranges demonstrated by current aerospace and industrial state of the art (Appendix C). The integrated mass envelope remains below 53% of commercial payload capacity even under pessimistic component bounds, indicating that the Pathfinder 3, Flying Whales LCA60T, and LZ 129 Hindenburg are strong candidate platforms for implementation on infrastructure analogous to existing industrial H_2 hubs.
8. Beyond the specific platforms evaluated, the closed-form sizing laws of Section 2.6 together with the consolidated subsystem framework of Tables 6 and 7 transfer directly to other heavy-lift configurations: any candidate platform reduces to a perturbation index ε and a structural mass, which the analytical machinery converts into actuator, condenser, and regeneration specifications. The regenerative cycle thus operates as a **generic actuation paradigm** for the next generation of heavy-lift airships, rather than as a configuration-specific result.

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Abbreviations

The following abbreviations and symbols are used in this manuscript:

Mathematical Symbols

A_{front}	Frontal (aerodynamic reference) area (m^2)
A_{total}	Total surface area of the airship (m^2)
C_D	Aerodynamic drag coefficient
D_{lim}	Maximum allowed preconditioning sinking depth (m)
$E_{H_2,\mathrm{el}}$	Specific energy demand for water electrolysis (kWh/kg)
F_{b}	Aerostatic buoyant thrust (N)
F_{drag}	Aerodynamic drag force (N)
f_{pan}	Photovoltaic surface coverage fraction
g	Acceleration due to gravity ($\mathrm{m/s}^2$)
h, Δh	Altitude and vertical displacement (m)
K_p, K_d, K_m	Control gains (proportional, derivative, and mass)
L_{eff}	Effective specific lift factor ($\rho_{\mathrm{air}}/\rho_{H_2} + 8$)
M_{str}	Structural mass (kg)
M_{total}	Total dynamic inertial mass (kg)
m_{H_2}	Mass of hydrogen gas inside the envelope (kg)
m_{load}	Mass of the payload to be released (kg)
m_{water}	Water ballast accumulated via O_2 absorption (kg)
$\dot{m}_{H_2}$	H_2 extraction/combustion rate (kg/s)
$\dot{m}_{H_2,\mathrm{min}}$	Minimum H_2 combustion rate for safe stabilization (kg/s)
$\dot{m}^*_{H_2}$	Minimax-optimal installed actuator capacity (kg/s)
$\dot{m}_{\mathrm{prec}}$	Proactive preconditioning flow rate (kg/s)
Q_{cond}	Condenser thermal power (kW)
r_{ef}	Effective solar yield ($\mathrm{kWh/m}^2\mathrm{/day}$)
T_1	Isothermal lifting gas temperature (K)
t_{anticip}	Anticipation time before payload release (s)
v	Vertical velocity (m/s)
V_{H_2}	Instantaneous volume of hydrogen lifting gas (m^3)
V_{eq}	Dynamic equilibrium H_2 volume (m^3)
α^*	Minimax-optimal proactive compensation fraction
ε	Perturbation index ($m_{\mathrm{load}}/M_{\mathrm{total}}$)
η_{el}	Electrolysis system efficiency
η_{gen}	Effective generator efficiency
ζ	Adaptive damping coefficient

Acronyms

COPV	Composite Overwrapped Pressure Vessel
HALE	High-Altitude Long-Endurance
PD	Proportional-Derivative
PEM	Proton Exchange Membrane
RFC	Regenerative Fuel Cell
URFC	Unitized Regenerative Fuel Cell

Appendix A Complete Derivations: Proactive Hydrogen Preconditioning

Appendix A.1 The Cubic Formula for Pre-Drop Sinking

Appendix A.1.1 Physical Setup and Mass Definition

Consider an airship at equilibrium altitude h_0 carrying a payload of mass m_{load}. The lifting gas volume is V_0 with density ρ_{H_2}. The ambient air density at altitude is ρ_{air}.

The total instantaneous inertial mass of the vehicle, M_{total}, includes the structural mass, the payload, and the mass of the hydrogen gas currently in the envelope:

$$M_{\mathrm{total}} = M_{\mathrm{str}} + m_{\mathrm{load}} + m_{H_2} \tag{A19}$$

At equilibrium (before payload release or gas extraction), the buoyant force exactly balances the total weight:

$$F_{\text{buoy},0} = \rho_{\text{air}} \cdot g \cdot V_0 = M_{\text{total}} \cdot g = W_0 \tag{A20}$$

Appendix A.1.2 Dynamics of Hydrogen Combustion and Water Retention

During the proactive preconditioning phase, hydrogen is transferred from the buoyancy envelope to the on-board generator at a constant mass flow rate $|\dot{m}|$. In the generator, this hydrogen reacts with atmospheric oxygen to produce liquid water, which is retained on board as ballast.

The stoichiometric ratio dictates that 1 kg of H_2 reacts with approximately 8 kg of atmospheric O_2 to produce 9 kg of H_2O. Therefore, while the envelope loses H_2 mass at a rate $|\dot{m}|$, the vehicle as a whole **gains** mass from the ingested oxygen at a rate $8 \cdot |\dot{m}|$.

The volume of the lifting gas as a function of time decreases linearly:

$$V_{H_2}(t) = V_0 - \frac{|\dot{m}|}{\rho_{H_2}} \cdot t \tag{A21}$$

The total weight of the airship as a function of time **increases**:

$$W(t) = (M_{\text{total}} + 8 \cdot |\dot{m}| \cdot t) \cdot g \tag{A22}$$

where M_{total} is the initial total mass at $t = 0$ (before preconditioning begins, so $m_{\text{water}}(0) = 0$). The increment $8|\dot{m}|t$ represents the net atmospheric O_2 mass absorbed; this is identically recovered in the dynamic simulation notation of Section 2 as $m_{\text{water}}(t) - |\dot{m}|t = 9|\dot{m}|t - |\dot{m}|t = 8|\dot{m}|t$, where $m_{\text{water}}(t) = 9|\dot{m}|t$ is the cumulative water ballast and $|\dot{m}|t$ is the H_2 mass removed from the envelope.

The net vertical force $\Delta F(t)$ acting on the airship is the difference between the instantaneous buoyancy and the instantaneous weight. *Remark*: Aerodynamic drag is neglected throughout the maneuver. During the preconditioning phase, drag opposes the downward velocity, reducing the actual sinking below Eq. (A27). During the post-release ascent, drag opposes the upward velocity, reducing the residual buoyancy excursion and therefore the reactive flow demand. In both phases the drag contribution is favorable; the drag-free model provides a conservative upper bound on actuator requirements in each phase independently.

$$\Delta F(t) = \rho_{\text{air}} \cdot g \cdot V_{H_2}(t) - W(t) \tag{A23}$$

Substituting the time-dependent expressions:

$$\Delta F(t) = \rho_{\text{air}} \cdot g \cdot \left(V_0 - \frac{|\dot{m}|}{\rho_{H_2}} \cdot t \right) - (M_{\text{total}} \cdot g + 8 \cdot |\dot{m}| \cdot g \cdot t) \tag{A24}$$

Since the initial equilibrium condition dictates that $\rho_{\text{air}} \cdot g \cdot V_0 = M_{\text{total}} \cdot g$, these terms cancel out:

$$\Delta F(t) = -\left(\frac{\rho_{\text{air}}}{\rho_{H_2}} + 8 \right) \cdot g \cdot |\dot{m}| \cdot t = -L_{\text{eff}} \cdot g \cdot |\dot{m}| \cdot t \tag{A25}$$

where $L_{\text{eff}} = \rho_{\text{air}}/\rho_{H_2} + 8$ is the effective specific lift factor introduced in Eq. (12).

Appendix A.1.3 The Cubic Formula

Applying Newton's second law, the vertical acceleration $a(t) \approx \Delta F(t)/M_{\text{total}}$ under the approximation $8|\dot{m}|t \ll M_{\text{total}}$ (valid for $S \equiv 8|\dot{m}|t/M_{\text{total}} < 0.3$, satisfied for typical operational timescales):

$$a(t) \approx -\frac{L_{\text{eff}} \cdot g \cdot |\dot{m}|}{M_{\text{total}}} \cdot t \tag{A26}$$

Integrating twice with initial conditions $v(0) = 0$, $\Delta h(0) = 0$:

$$\boxed{|\Delta h(t)| = \frac{L_{\text{eff}}\, g\, |\dot{m}|}{6\, M_{\text{total}}}\, t^3} \tag{A27}$$

The cubic dependence arises because a constant flow rate produces a linear reduction in gas volume and a linear increase in ballast mass, both creating a linear growth in net downward force. Integrating this linear force twice yields a cubic position trajectory.

Appendix A.2 Maximum Allowed Anticipation Time

Given that the airship cannot sink more than D_{max} before the payload is released, and defining $|\dot{m}| = \eta \cdot \dot{m}_{\text{max}}$, setting $|\Delta h(t_{\text{max}})| = D_{\text{max}}$ and solving:

$$\boxed{t_{\text{max}}(\eta, D_{\text{max}}) = \left(\frac{6 \cdot M_{\text{total}} \cdot D_{\text{max}}}{L_{\text{eff}} \cdot g \cdot \eta \cdot \dot{m}_{\text{max}}}\right)^{1/3}} \tag{A28}$$

Appendix A.3 Optimal Sinking Distance for Perfect Compensation

Perfect compensation ($\alpha = 100\%$) occurs when the net downward force generated during preconditioning exactly equals the weight of the payload to be released. The mass of hydrogen that must be burned to neutralize a payload m_{load} is:

$$\Delta m^*_{\text{pre}} = \frac{m_{\text{load}}}{L_{\text{eff}}} \tag{A29}$$

Setting $t_{\text{anticip}} = \tau \cdot t_{\text{max}}$ and equating to Δm^*_{pre}, then cubing and solving for D^*_{max}:

$$\boxed{D^*_{\text{max}} = \frac{m^3_{\text{load}} \cdot g}{6 \cdot M_{\text{total}} \cdot L^2_{\text{eff}} \cdot \eta^2 \cdot \dot{m}^2_{\text{max}} \cdot \tau^3}} \tag{A30}$$

Appendix A.4 Compensation Fraction as Function of Sinking

The compensation fraction α is defined as the mass actually burned divided by the mass required for perfect compensation:

$$\alpha = \frac{\Delta m_{\text{pre}}}{m_{\text{load}}/L_{\text{eff}}} \tag{A31}$$

Dividing the expression for Δm_{pre} at arbitrary depth D_{max} by the expression at perfect compensation D^*_{max}, all parameters except the distances cancel:

$$\boxed{\alpha(D_{\text{max}}) = \left(\frac{D_{\text{max}}}{D^*_{\text{max}}}\right)^{1/3}} \tag{A32}$$

Appendix A.5 Minimum Flow Requirement for Viability

If the operational sinking is limited to D_{limit} and a target compensation α_{target} is required, substituting $D_{\text{limit}} = \alpha^3_{\text{target}} \cdot D^*_{\text{max}}$ into the expression for D^*_{max} and solving for $\dot{m}_{\text{max}}$:

$$\boxed{\dot{m}_{\text{min}} = \frac{m_{\text{load}}^{3/2} \cdot g^{1/2} \cdot \alpha_{\text{target}}^{3/2}}{\sqrt{6 \cdot M_{\text{total}} \cdot L_{\text{eff}}^2 \cdot D_{\text{limit}} \cdot \eta \cdot \tau^{3/2}}}} \tag{A33}$$

For perfect compensation ($\alpha_{\text{target}} = 1$):

$$\boxed{\dot{m}_{\text{min}}^{\text{pre-comp}} = \frac{m_{\text{load}}^{3/2} \cdot \sqrt{g}}{\eta \cdot \tau^{3/2} \cdot \sqrt{6 \cdot M_{\text{total}} \cdot L_{\text{eff}}^2 \cdot D_{\text{limit}}}}} \tag{A34}$$

Key result: for a fixed operational sinking limit D_{limit}, the minimum required flow scales inversely with the square root of the total inertial mass, recovering the scaling paradox of Eq. (11):

$$\dot{m}_{\text{min}} \propto \frac{m_{\text{load}}^{3/2}}{\sqrt{M_{\text{total}}}} \tag{A35}$$

Appendix A.6 System-Level Actuator Optimization (Minimax Formulation)

Since both the proactive preconditioning and the reactive post-drop stabilization share the same combustion hardware, the installed capacity $\dot{m}_{\text{max}}$ must satisfy the peak demand of the more intensive phase. From Appendix A.5, the proactive demand scales as $\dot{m}_{\text{pre}}(\alpha) = K_{\text{pre}} \cdot \alpha^{3/2}$ (monotonically increasing), while the reactive demand scales as $\dot{m}_{\text{post}}(\alpha) = \dot{m}_{\text{post},0} \cdot (1 - \alpha)$ (monotonically decreasing). The optimal installed capacity minimizes the peak of both:

$$\dot{m}_{H_2}^{*} = \min_{\alpha \in [0,1]} \Big(\max\big(\dot{m}_{\text{pre}}(\alpha),\, \dot{m}_{\text{post}}(\alpha)\big) \Big) \tag{A36}$$

The minimum occurs at the intersection $\dot{m}_{\text{pre}}(\alpha^*) = \dot{m}_{\text{post}}(\alpha^*)$. Introducing the non-dimensional system constant:

$$C = \frac{\dot{m}_{\text{post},0} \cdot \eta \cdot \tau^{3/2} \cdot \sqrt{6 \cdot M_{\text{total}} \cdot L_{\text{eff}}^2 \cdot D_{\text{limit}}}}{m_{\text{load}}^{3/2} \cdot \sqrt{g}} \tag{A37}$$

the change of variables $x = \sqrt{\alpha^*}$ transforms the fixed-point equation into the depressed cubic $x^3 + Cx^2 - C = 0$, which has exactly one strictly positive real root by Descartes' Rule of Signs. The exact closed-form solution depends on the discriminant $\Delta = C^2/4 - C^4/27$:

High-inertia regime ($\Delta > 0$, i.e., $C < 3\sqrt{3}/2$): unique real root via Cardano's method:

$$\alpha^* = \left(\sqrt[3]{\frac{C}{2} - \frac{C^3}{27} + \frac{C}{2}\sqrt{1 - \frac{4C^2}{27}}} + \sqrt[3]{\frac{C}{2} - \frac{C^3}{27} - \frac{C}{2}\sqrt{1 - \frac{4C^2}{27}}} - \frac{C}{3} \right)^2 \tag{A38}$$

Low-inertia regime ($\Delta \leq 0$, i.e., $C \geq 3\sqrt{3}/2$): three real roots (*Casus Irreducibilis*); the unique physical root via Vieta's trigonometric substitution ($k = 0$):

$$\alpha^* = \left(\frac{2C}{3} \cos\!\left(\frac{1}{3} \arccos\!\left(\frac{27 - 2C^2}{2C^2} \right) \right) - \frac{C}{3} \right)^2 \tag{A39}$$

These closed-form expressions eliminate the need for numerical root-finding in the actuator sizing phase and directly yield the minimax-optimal preconditioning fraction α^* and installed capacity $\dot{m}_{H_2}^{*}$ for any given airship geometry and payload configuration.

Appendix B Adaptive PD Controller: Full Implementation

Gain Scheduling

The derivative gain is adaptively tuned to maintain a constant damping coefficient $\zeta = 1.35$ throughout the unloading sequence:

$$K_d(t) = 2\,\zeta\,\sqrt{\frac{K_p\,M_{\text{eff}}(t)}{\rho_{\text{air}}(h)\,g}}, \qquad M_{\text{eff}}(t) = M_{\text{str}} + M_{\text{load}}(t) + m_{\text{water}}(t) \tag{A40}$$

where M_{eff} includes the structural mass, remaining payload, and accumulated water ballast, but excludes m_{H_2} (gas mass contributes to buoyancy rather than inertial damping). This expression is derived by imposing that the damping coefficient ζ of the linearized system remains constant under mass variations; without this adaptation, the step-wise loss of payload and concurrent gain of water ballast would progressively alter the dynamic regime.

Feedback Flow Law

The mass flow rate commanded by the feedback controller is:

$$\dot{m}_{\text{PD}}(t) = K_m\left(V_{\text{eq}}(t) + K_p\,e_h - K_d(t)\,v - V_{H_2}\right) \tag{A41}$$

where $e_h = h_0 - h$ is the altitude error. The gain K_m operates in two sequential phases: an *emergency phase* ($K_m = 60$ kg/s/m^3) provides rapid post-drop response until m_{H_2} reaches the new equilibrium target $m_{H_2,\text{eq}} = M_{\text{eff}}/(\rho_0/\rho_{H_2} - 1)$; a *trim phase* ($K_m = 0.5$ kg/s/m^3) then guides smooth settling to the new equilibrium. The emergency phase ensures aggressive compensation of the initial perturbation; the reduced trim gain prevents oscillation buildup and resets automatically at each subsequent drop event.

Anticipatory Preconditioning and Hybrid Control Law

To mitigate the initial acceleration peak, the system incorporates an anticipatory extraction pulse prior to the scheduled release instant (t_{drop}). As detailed in Section 2, extracting a negative flow $\dot{m}_{\text{prec}}$ during the anticipation window t_{anticip} induces a proactive lift deficit that partially absorbs the perturbation before release, strictly maintaining the preconditioning sinking depth $|\Delta h(t_{\text{anticip}})|$ below the operational bound D_{lim}.

The overall hybrid control law, incorporating both the anticipatory pulse and the saturated PD feedback, is defined as:

$$\dot{m}_{H_2}(t) = \begin{cases} \dot{m}_{\text{prec}} & \text{if } t \in [t_{\text{drop}} - t_{\text{anticip}},\ t_{\text{drop}}) \\ \text{sat}_{\dot{m}_{H_2,\max}}(\dot{m}_{\text{PD}}(t)) & \text{otherwise} \end{cases} \tag{A42}$$

This saturation at $\pm\dot{m}_{H_2,\max}$ is the critical design parameter. The proactive pulse qualitatively transforms the actuator requirement: instead of forcing the system to compensate for an already-initiated upward impulse, the PD controller engages a vehicle that is already partially in descent, drastically reducing the reactive flow required.

Appendix C Expanded Hardware Validation Material

This appendix expands the subsystem-level data behind Tables 6 and 7 in Section 4.6. Bounds are reported in two regimes throughout: the *industrial conservative* regime (commercial off-the-shelf systems) and the *aerospace SOTA 2025* regime (peer-reviewed aviation-grade systems). When the regimes diverge, the per-platform mass envelope of Table 7 reflects the aerospace SOTA bound.

Appendix C.1 PEM Electrolyser — Mass Breakdown

The PEM electrolyser sustains 44.7 kg H_2/day at ≈84 kW continuous. Table A8 brackets the system mass across three aerospace technology bounds; the NREL industrial cost-modelling envelope (Mayyas 2019) of 100–420 kg frames the upper reference, well above all aerospace SOTA 2025 values.

Table A8. PEM electrolyser mass bounds at 84 kW continuous power (system mass at $\eta = 65$–80%).

Bound	Mass (kg)	Power density (kW/kg)	Technology / Reference
Optimistic	∼50	2.2	HyPoint HT-PEM, aviation-grade (incl. min. BoP 6–22.7%) [30,39]
Central	∼85	1.0–1.2	Water-cooled mainstream (incl. BoP) [30, 40]
Pessimistic	∼170	0.5	Air-cooled PEM [30]
Theoretical	∼30	2.77	NASA Glenn bare stack at 84 kW [38]

Appendix C.2 Generator Unit — Modular Burst-Duty Architecture

The module size of ∼1 kg/s is hardware-constrained: Cameretti et al. demonstrated 100% H_2 combustion in a GE90-class chamber at 121–150 MW thermal [31]; at the H_2 lower heating value of 120 MJ/kg, this corresponds to 1.0–1.25 kg/s per chamber (conservative bound: 1 kg/s; per-module thermal output: 120 MW). Table A9 summarises the module mass bounds; the industrial option is excluded as incompatible with burst-duty operation.

Table A9. Generator module mass bounds (burst-duty ∼30 s/cycle; per-module thermal power 120 MW).

Bound	Module mass (kg/module)	Technology basis / Reference
Optimistic	∼250	APU-class burst-duty: Garrett GTCP 36-300 with micromix combustor (1.6 MW+370 kW electric, ≈150–180 kg) [37]
Pessimistic	∼600	Turbofan continuous-duty fraction: GE90 combustor section ($M_{eng} = 8{,}283$ kg total)
Industrial (excl.)	∼1,800	Stationary turbines (e.g. Wärtsilä, MAN) at ∼15 t/MW thermal — burst-duty incompatible

The number of parallel modules per platform is $N = \lceil \dot{m}^*_{H_2}/1\,\mathrm{kg/s} \rceil$: $N = 4$ (LZ 129 Hindenburg), $N = 5$ (Flying Whales LCA60T), $N = 6$ (Pathfinder 3), $N = 3$ (Zeppelin NT, multi-drop 4×250 kg).

Appendix C.3 Photovoltaic Array — Per-Platform Mass

Per-platform panel masses follow from the thin-film amorphous baseline validated for airships [32]: surface density 120 g/m^2 at $\eta = 8\%$, applied to $f_{pan} = 0.45$ of the upper-surface area of each platform (Table A10).

Table A10. Photovoltaic array effective area and mass per platform (thin-film amorphous baseline).

Platform	Effective PV area (m^2)	Array mass (kg)
LZ 129 Hindenburg	10,800	1,296
Flying Whales LCA60T	10,350	1,242
Pathfinder 3	5,850	702
Zeppelin NT	1,260	151

Substituting single-junction (253 g/m^2, $\eta = 24\%$) or triple-junction (345 g/m^2, $\eta = 32\%$) cells scales the array mass by $\sim 2.1\times$ or $\sim 2.9\times$ respectively and reduces the regeneration cycle duration in inverse proportion to η.

Appendix C.4 COPV Storage — Per-Platform Mass

Type-IV composite overwrapped pressure vessels (COPV) at 350 bar achieve 5.5 wt% gravimetric density (FCEV-grade, DOE/EERE; SAE J2601-2 [17]), corresponding to 17.2 kg tank per kg of stored H_2. The per-cycle H_2 reserve is obtained from the simulation logs (including transient margin) and dictates the tank sizing shown in Table A11:

Table A11. COPV storage mass derived from per-cycle H_2 reserve (simulation logs).

Platform	H_2 reserve (kg/cycle)	COPV mass (kg)
LZ 129 Hindenburg	96.7	1,663
Flying Whales LCA60T	104.8	1,802
Pathfinder 3	28.9	497
Zeppelin NT	33.7	580

Appendix C.5 H_2 Compressor — Mass not Reported in Open Literature

The diaphragm compressor (30–40 bar electrolyser output to 350 bar storage) operates at $\sim$1.86 kg/h flow and $\sim$6 kW input power. Commercial units in this performance envelope include the Hofer Case 2 (1.8 kg/h, 0.8 $\rightarrow$ 80.1 MPa, 11 kW, 2-stage diaphragm; [45]) and the PDC Machines C-series [33]. Due to the proprietary nature of commercial compressor datasheets, mass specifications were estimated using comparable industrial-scale references [45]. The mass budget applies an engineering estimate of $\sim$100 kg.

Appendix C.6 Envelope and Ballonet Material Compatibility with Hydrogen

Gaseous H_2 at low positive pressures ($\Delta P = 1$–50 mbar) imposes specific material requirements on gas cell and ballonet membranes, owing to hydrogen's small molecular size and consequent permeability through standard flexible materials. The trilayer laminate construction validated on the Zeppelin NT [36,48] addresses these requirements through a dedicated functional architecture: an outer Tedlar (PVF) film provides UV resistance and environmental protection; a load-bearing woven polyester fabric carries the hoop stresses from the internal pressure margin; and an inner polyurethane (PU) coating constitutes the primary H_2 containment barrier, combining low gas permeability with thermal weldability for continuous seam integrity. This construction has been operational in H_2-lift airship service and constitutes the material reference for the ballonet system in this study.

Appendix C.7 Note on Regenerative Fuel Cell Consolidation

The model treats the electrolyser and the H_2 combustion generator as independent subsystems. Aerospace regenerative fuel-cell (RFC) systems, demonstrated onboard the NASA Helios HALE prototype with energy densities of $\sim$450 Wh/kg and 18.5 kW power output [35], integrate both functions in a unified stack with round-trip efficiencies of 33–55% (1990s alkaline, 2003 PEM). An RFC architecture would consolidate the electrolyser and the combustion-derived electrical recovery into a single subsystem, trading mass-budget simplification for additional integration complexity — a direction left to future work.

Appendix C.8 Consolidated Mass Budget per Subsystem and Platform

Table A12 consolidates the contributions of Appendices C.1–C.5 into a per-platform, per-subsystem mass budget. Optimistic and pessimistic bounds are taken from the individual sections; their column sums reproduce the totals of Table 7 in Section 4.6.

Table A12. Per-subsystem, per-platform mass contributions (kg). Optimistic (opt) and pessimistic (pes) bounds; column sums reproduce Table 7.

	Hindenburg		Flying Whales		Pathfinder 3		Zeppelin NT	
Subsystem	opt	pes	opt	pes	opt	pes	opt	pes
PEM electrolyser	50	170	50	170	50	170	50	170
Generator unit (*N* modules)	1,000	2,400	1,250	3,000	1,500	3,600	750	1,800
Photovoltaic array	1,296	1,296	1,242	1,242	702	702	151	151
COPV storage	1,663	1,663	1,802	1,802	497	497	580	580
Condenser	50	50	50	50	50	50	50	50
H_2 compressor	100	100	100	100	100	100	100	100
Water tank	50	50	50	50	50	50	50	50
Ballonets & piping	80	80	80	80	80	80	80	80
Manifold & minor systems	20	20	20	20	20	20	20	20
Total	**4,309**	**5,829**	**4,644**	**6,514**	**3,049**	**5,269**	**1,831**	**3,001**

The number of generator modules is $N = 4, 5, 6, 3$ for the LZ 129 Hindenburg, Flying Whales LCA60T, Pathfinder 3, and Zeppelin NT (multi-drop 4×250 kg) respectively, derived from $N = \dot{m}^{*}_{H_2}/1\,\mathrm{kg/s}$ (Appendix C.2). Only the PEM electrolyser and the generator unit are reported with two bounds; the remaining subsystems are treated as single-valued contributions in this consolidation, with their individual ranges discussed in Appendices C.3–C.5 and in the narrative of Section 4.6 (condenser 30–76 kg range; compressor $\sim$100 kg engineering estimate).

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